

Project Title: **Design of a Double Reduction Spur Gear Speed Reducer for a Commercial Tractor**

Course: ME 154 Lab

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Link for the presentation: [Final Project - ME 154.pptx](#)

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1. Introduction

1.1 Project Overview

In this design project, we are to calculate, design, and implement a geartrain to support a commercial tractor that works against snow removal or lawn mowing. The components of this geartrain consist of spur gears, shafts, bearings, housing, and fasteners. The geartrain costs are to be kept to a minimum while meeting specifications. Specifications include an infinite life for the shaft and a minimum of 5,000 operating hours for the gears and bearings. The gasoline engine of the tractor will supply 6 horsepower to the geartrain, where the input pinion revolves at 2000 revolutions per minute, and the output gear will revolve at 185 revolutions per minute. To achieve this, the implementation of a speed reducer is necessary. This speed reducer is a step shaft that includes 2 gears, and through it, we lower the revolutions per minute while increasing the torque of the output gear shaft.

1.2 Design Objectives

The design of the geartrain must adhere to several specifications. The input pinion will revolve at 2000 revolutions per minute and the output gear is to revolve at 185 revolutions per minute. The torque the output shaft produces should be enough to drive the wheels of the tractor. Ultimately, the geartrain is to be durable, ensuring an infinite life of the shaft and a minimum of 5,000 operational hours on the spur gears and bearings. The geartrain is to operate safely, ensuring operator safety. The geartrain size is to be kept to a minimum. Altogether being compact and low cost.

1.3 Design Requirements

In this report, you can expect an analysis of a double reduction spur gear train components. These components include gear train housing, spur gears, shafts, bearings, and fasteners. Each of these components will come with supplemental CAD assembly, FEA validation, MATLAB calculations, and animation video(s).

2. Design Methodology

2.1 Assumptions

For performing calculations on the geartrain, we must assume a few things in order to achieve theoretical values. The first assumption of this analysis is steady state loading. Torque is considered to be constant during operations. We assume that there is uniform power transmission neglecting any fluctuations or vibrations. We analyse our geartrain only using spur gears, avoiding the usage of helical gears and assuming the spur gears are manufactured to the tightest tolerance, with no interference to account for. We use the standard pressure angle (ϕ) of 20° while operating in a room temperature environment. Finally, we assume that the engine is supplying a constant revolution per minute of the shaft to the pinion powering the gear train.

Through these assumptions, we are able to achieve theoretical values but neglect impact forces, vibrations, dynamic loadings and so on.

2.2 Design Flowchart

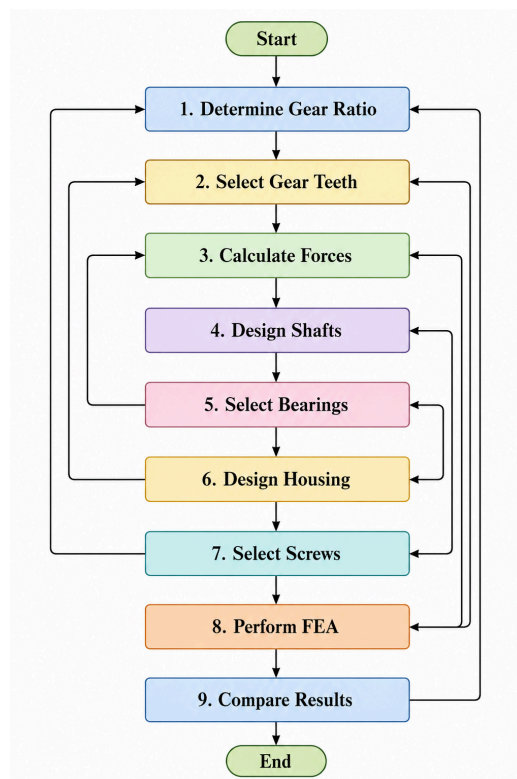


Figure 1. Flowchart

3. Gear Train Design

3.1 Overall Gear Ratio

To select a gear ratio, we used the desired revolutions per minute of the pinion and gear. The result of the ratio was significant, and part of creating the gear train is to make it compact and cost-effective; therefore, we square-rooted the ratio to keep the ratio and minimize the size of our gears. The inclusion of this step reduced the ratio by 30%. Without this step, the pinion would have to have been larger, and this is unfeasible for the geartrain due to the sizing and cost constraints our team is facing.

We opted for a double reduction gear system because of the geartrain's objectives. We are to take a large revolution per minute and a low-torque pinion and convert it to a significantly smaller revolution per minute and a large torque gear. The double reduction gear system does exactly that. To add on, the durability of gears is increased as the stresses are spread among two stages. The geartrain can be expected to operate smoothly with reasonable reductions and reduced stress application.

3.2 Gear Ratio Distribution

$$n_p = 2000 \text{ RPM} \quad n_g = 180 \text{ RPM}$$

$$m = \frac{2000}{180} = 11.11$$

$$\sqrt{11.11} = 3.33 \Rightarrow 3.33:1$$

3.3 Gear Material Selection

For our gears, we ended up selecting CD AISI 1045. The Brinell Hardness of the material is 250, with a $S_y = 77$ kpsi and $S_{uts} = 91$ kpsi. We selected AISI 1045 due to the constraints we are working under. We need a strong material that is cost effective. AISI 1045 is stronger and cheaper than alloy steels. The material is easily manufactureable which plays a role in the cost effectiveness of the material. The hardness of the material resists wear on gear teeth while the toughness is greater being less brittle than steels with larger carbon contents.

As a team we selected the cold drawn treatment so the tolerances are tighter and the finish is better allowing for accurate teeth geometry and smoother meshing. Ultimately, with the gears we would like to avoid any form of interference as it would affect the integrity of the gears.

3.4 Gear Geometry

$$m = 3.33 \quad k = 1 \text{ (Full Depth Teeth)} \quad \phi = 20^\circ$$

$$N_{P(Min)} = \frac{2k}{(1+2m)\sin^2(\phi)} [m + \sqrt{m^2 + (1+2m)\sin^2(\phi)}] = 15.6 = 16 \Rightarrow 30 \text{ Teeth}$$

$$N_g = N_p(m) = 99.9 = 100 \text{ Teeth}$$

$$m_{true} = \frac{100}{30} = 3.33 (\pm 0\%)$$

$$C2C = \left(\frac{d_p}{2}\right) + \left(\frac{d_g}{2}\right) = 6.5"$$

$$\frac{\omega_p}{\omega_g} = \frac{N_g}{N_p} = \frac{d_g}{d_p} = \frac{r_g}{r_p} = m$$

$$\Rightarrow d_g = m(d_p) = 3.33(3.0) = 10.0"$$

$$\Rightarrow \left(\frac{d_p}{2}\right) + \left(\frac{m(d_p)}{2}\right) = C2C \Rightarrow d_p = \frac{2(C2C)}{1+m} = 3.0"$$

$$r_g = 5.0" \quad r_p = 1.5"$$

$$P = \frac{N}{d} = 10 \frac{\text{Teeth}}{\text{in}}$$

$$p \times P = \pi \Rightarrow p = \frac{\pi}{P} = \frac{\pi}{10}$$

$$3\left(\frac{\pi}{10}\right) < F < 5\left(\frac{\pi}{10}\right) = 0.942" < F < 1.571" \Rightarrow F = 1.25"$$

$$\text{Addendum} = \frac{1}{P} = 0.1"$$

$$\text{Dedendum} = \frac{1.25}{P} = 0.125"$$

3.5 Gear Force Calculations

$$H = 6 \text{ hp}$$

$$v = \frac{\pi d n}{12}$$

$$\Rightarrow v_{GS1} = 1570.8 \frac{ft}{min}$$

$$\Rightarrow v_{GS2} = 471.2 \frac{ft}{min}$$

$$W^t = 33000 \frac{H}{v}$$

$$\Rightarrow W_{23}^t = 126.05 \text{ lb} = F_a^t$$

$$\Rightarrow W_{45}^t = 420.17 \text{ lb} = F_d^t$$

$$F_{23}^n = \frac{F_{23}^t}{\cos(20^\circ)} = 134.14 \text{ lb}$$

$$F_{23}^r = F_{23}^n \sin(20^\circ) = 45.88 \text{ lb}$$

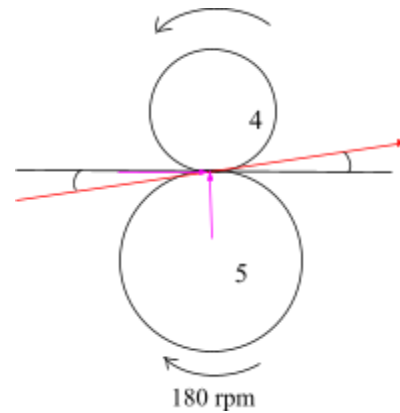
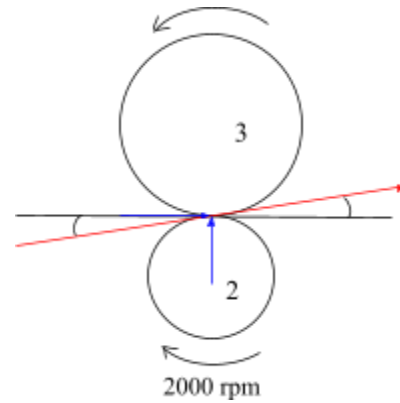
$$F_{45}^n = \frac{45^t}{\cos(20^\circ)} = 447.13 \text{ lb}$$

$$F_{45}^r = F_{45}^n \sin(20^\circ) = 152.93 \text{ lb}$$

$$HP = \frac{T \cdot n}{5252} \Rightarrow T = \frac{HP(5252)}{n}$$

$$T_p = \frac{6(5252)}{2000} = 15.76 \text{ lb ft} = 189.07 \text{ lb in}$$

$$T_g = \frac{6(5252)}{180} = 175.07 \text{ lb ft} = 2100.8 \text{ lb in}$$



3.6 AGMA Stress Analysis

DIFFERENT GEARSETS FOLLOW THE SAME ANALYSIS. CALCULATIONS DONE THROUGH MATLAB.

Bending Stress

PINION

$$W^t = 126.05 \text{ lb}$$

$$v_p = 1570.8 \frac{ft}{min}$$

$$Q_v = 6$$

$$J = 0.39 \text{ (Fig. 14 - 6)}$$

$$k_v = \left(\frac{A+\sqrt{V}}{A}\right)^B = 1.522$$

$$A = 50 + 56(1 - B) = 59.773$$

$$B = 0.25(12 - Q_v)^{\frac{2}{3}} = 0.826$$

$$\sigma_{B_p} = W^t k_o k_v k_s \frac{P}{F} \frac{k_m k_b}{J} = (126.05)(1)(1.522)(1)\left(\frac{10}{1.25}\right)\left(\frac{1.6(1)}{0.39}\right) = 6295.74 \text{ psi}$$

Contact Stress

PINION

$$m = 3.33$$

$$\phi = 20^\circ$$

$$C_p = 2100 \text{ (Table 14 - 8)}$$

$$C_f = 1$$

$$I = \frac{\cos(\phi_t)\sin(\phi_t)}{2m_n} \frac{M_g}{M_g + 1} \text{ (External Gear)} = 0.1236$$

$$\sigma_c = C_p \left(W^t k_o k_v k_s \frac{k_m}{d_p F} \frac{C_f}{I} \right)^{\frac{1}{2}} = (2100) \left((126.05)(1)(1.522)(1) \left(\frac{1.6}{10 \cdot 1.25} \right) \left(\frac{1}{0.1236} \right) \right)^{\frac{1}{2}} = 54035.88 \text{ psi}$$

Safety Factor

PINION

BENDING

$$N = 60 \frac{\text{min}}{\text{hour}} \times (5000 \text{ hours}) \times 2000 \frac{\text{rev}}{\text{min}} = 600(10)^6 \text{ cycles}$$

$$Y_N = 1.3558(N)^{-0.0178} = 0.946$$

$$K_T = 1 \text{ (} T < 250^\circ F \text{)}$$

$$K_R = 1 \text{ (Reliability 99\% - Table 14.10)}$$

$$S_t = 77.3H_B + 12800 \text{ (Grade 1)} = 77.3(250) + 12800 = 32125 \text{ psi}$$

$$S_f = \frac{\left(\frac{S_{tN}}{K_T K_R} \right)}{\sigma} = \frac{(32125)(0.946)}{6295.74} = 4.83$$

CONTACT

$$S_C = 322H_B + 29100 = 322(250) + 29100 = 109600 \text{ psi}$$

$$Z_N = 1.4488N^{-0.023} = 1.4488(600(10)^6)^{-0.023} = 0.910$$

$$S_H = \frac{\frac{S_C Z_N}{K_T K_R}}{\sigma_C} = \frac{109600(0.910)}{54035.88} = 1.85$$

3.7 Gear Design Summary Table

Table 1. Gear Design Parameters

Variable	Symbol	Value	Unit
Pinion Teeth	N_p	30	teeth
Gear Teeth	N_g	100	teeth
Pinion pitch radius	r_p	1.5	inches
Gear pitch radius	r_g	5	inches
Center to Center Distance	C2C	6.5	inches
Diametral Pitch	P	10	$\frac{\text{teeth}}{\text{in}}$
Face Width	F	1.25	inches
Addendum	-	0.1	inches
Dedendum	-	0.125	inches

4. Shaft Design

4.1 Shaft Geometry

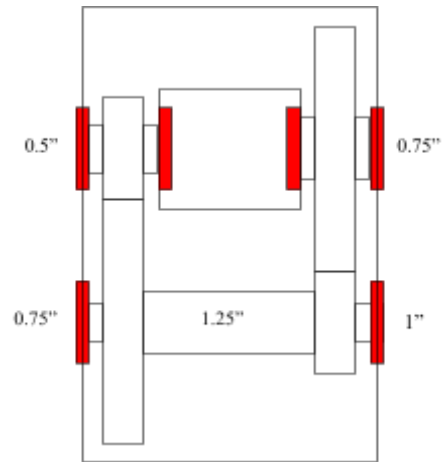


Figure 2. Shaft Geometry Top View

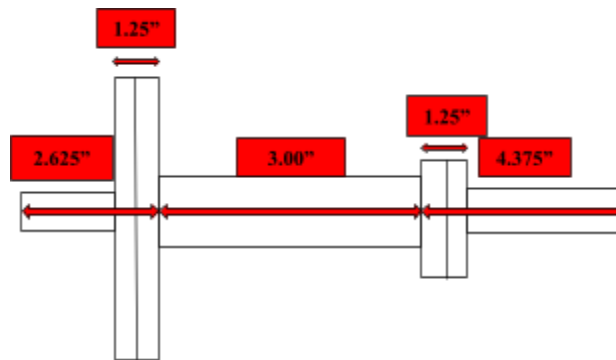


Figure 3. Step Shaft Geometry

4.2 Free Body Diagrams

Step Shaft Analysis:

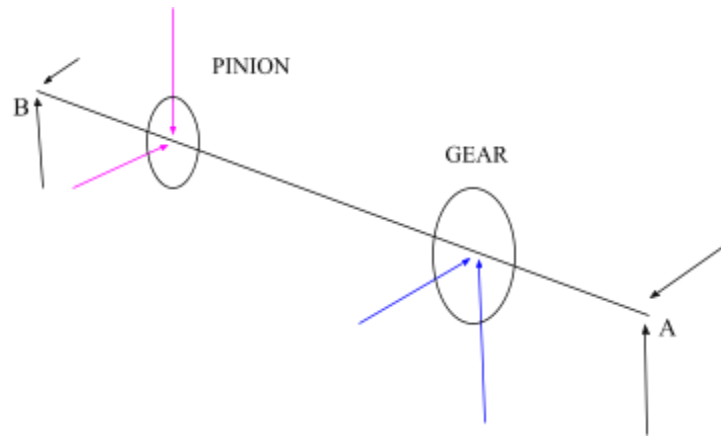


Figure 4. Reaction Force Analysis

$$\Sigma F_y = 0 \Rightarrow R_A^y + R_B^y = F_g^r + F_p^r \Rightarrow R_A^y = 94.1 \text{ lbf}$$

$$\Sigma M_A = - (F_g^r)(2'') - (F_p^r)(6.25'') + (R_B^y)(10'') \Rightarrow R_B^y = 104.8 \text{ lbf}$$

$$\Sigma F_z = 0 \Rightarrow R_A^z + R_B^z = F_a^t + F_d^t = 0 \Rightarrow R_A^z = 56.7 \text{ lbf}$$

$$\Sigma M_A = - (F_a^t)(4.75'') - (F_d^t)(9'') + (R_B^z) = 0 \Rightarrow R_B^z = 237.4 \text{ lbf}$$

$$R_B = \sqrt{(R_B^y)^2 + (R_B^z)^2} = 259.5 \text{ lbf} = R_p$$

$$R_A = \sqrt{(R_A^y)^2 + (R_A^z)^2} = 109.83 \text{ lbf} = R_g$$

4.3 Shear & Moment Diagrams

STEP SHAFT

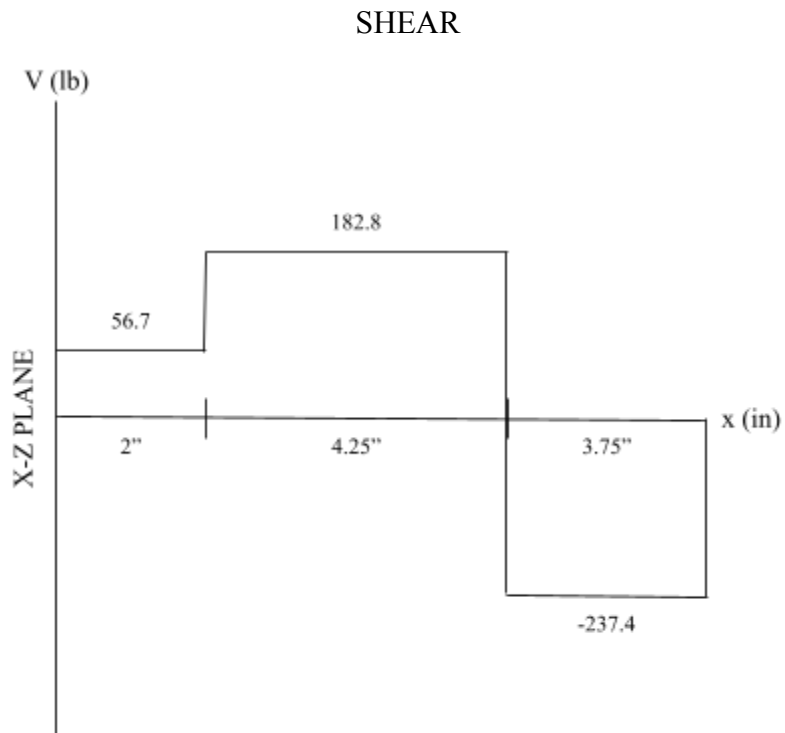


Figure 5. Step Shaft Shear Diagram (X-Z)

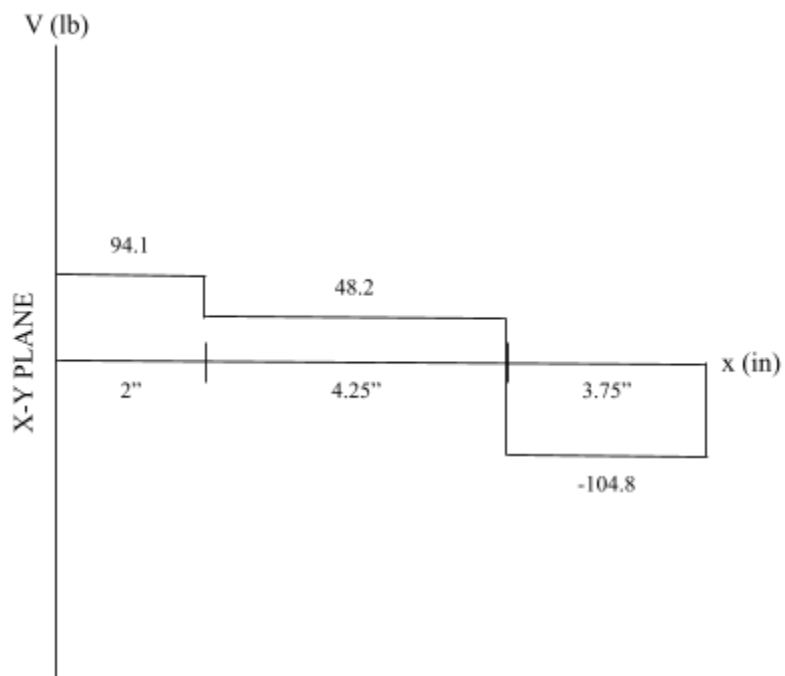


Figure 6. Step Shaft Shear Diagram (X-Y)

MOMENT

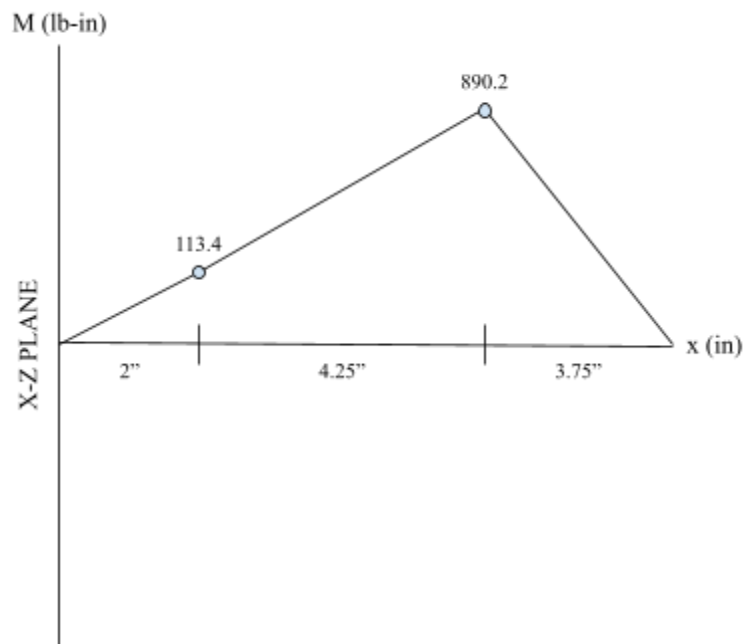


Figure 7. Step Shaft Moment Diagram (X-Z)

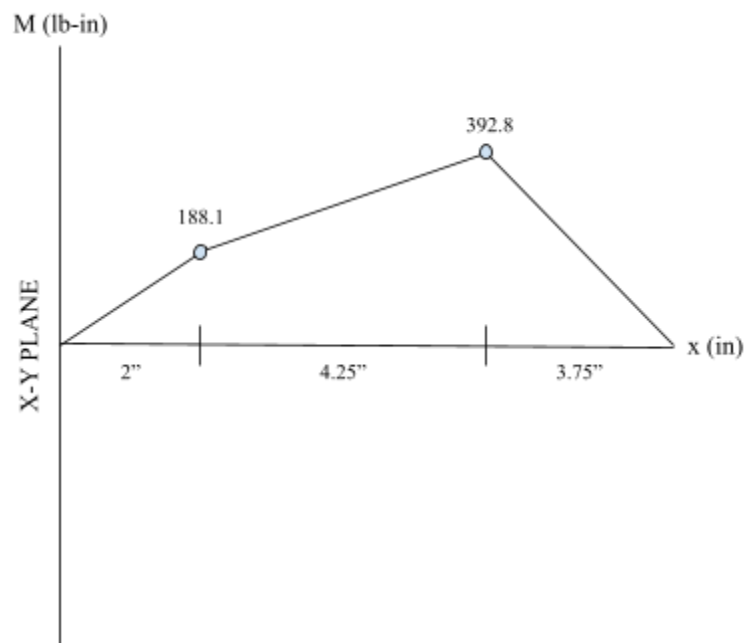


Figure 8. Step Shaft Moment Diagram (X-Y)

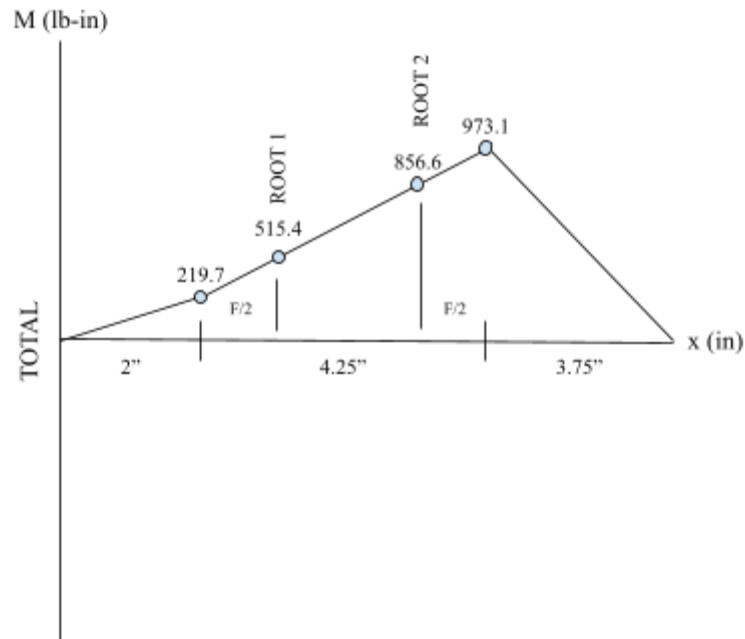


Figure 9. Step Shaft Total Moment Diagram

PINION & GEAR SHAFT(S)

Since the gear and pinion shafts are simply supported and symmetrical, we can divide the resultant force by 2 to of each to get the maximum shear, then solve for their moments.

$$F_{23}^n = 134.14 \text{ lb} \Rightarrow R_1 = 67.07 \text{ lbf}$$

$$F_{45}^n = 447.13 \text{ lb} \Rightarrow R_3 = 223.56 \text{ lbf}$$

4.4 Shaft Diameter Calculations

Fatigue Theory: DE Goodman Criteria

$$S'_e = S_{ut}(0.5) = 45.5 \text{ kpsi} (S_{ut} < 200 \text{ kpsi})$$

$$k_a = 2(S_{ut})^{-0.217} = 0.751 \text{ (Surface Factor)}$$

$$k_b = \left(\frac{d}{0.3}\right)^{-0.107} = 0.879 \text{ (Size Factor)}$$

$$k_c = 1 \text{ (Loading Factor – Bending)}$$

$$k_d = 1 \text{ (Temperature Factor)}$$

$$k_e = 0.814 \text{ (Reliability Factor – 99\%)}$$

$$S_e = S_e'(k_a)(k_b)(k_c)(k_d)(k_e) = 24.4 \text{ kpsi}$$

$$K_f = 1 + q(K_t - 1)$$

$$K_{fs} = 1 + q(K_{ts} - 1)$$

$$A = \sqrt{4(K_f M)^2}$$

$$B = \sqrt{3(K_{fs} T)^2}$$

$$d = \frac{16n}{\pi} \left[\frac{A}{S_e} + \frac{B}{S_{ut}} \right]^{\frac{1}{3}}$$

Through the MATLAB code, we reiterate the process until we get a final d value, then apply the relations:

$$d = 1"$$

$$D = 1.2d = 1.25" \text{ (Rounded to the nearest } \frac{1}{8} ")$$

$$d_1 = 0.75"$$

$$D_1 = \frac{5}{3}(d_1) = 1.25"$$

4.5 Stress Analysis

$$\sigma = \frac{32(K_f)(M)}{\pi(d^3)}$$

$$\tau = \frac{16(K_{fs})(T)}{\pi(d^3)}$$

Results:

Countershaft Results

$$(\sigma)_{Fillet1} = 981.7 \text{ psi}$$

$$(\sigma)_{Fillet2} = 2581.4 \text{ psi}$$

$$(\sigma)_{Pinion} = 1946.1 \text{ psi}$$

$$(\sigma_m)_{Fillet1} = 1484.6 \text{ psi}$$

$$(\sigma_m)_{Fillet2} = 1418.6 \text{ psi}$$

$$(\sigma_m)_{Pinion} = 1091.6 \text{ psi}$$

$$(n_y)_{Fillet2} = 1.5$$

$$(n_y)_{Fillet2} = 1.6$$

$$(n_y)_{Pinion} = 2.1$$

Input Shaft

$$(\sigma_a) = 268.3 \text{ psi}$$

$$(\sigma_m) = 327.5 \text{ psi}$$

$$(n_y) = 1.8$$

Output Shaft

$$(\sigma_a) = 894.3 \text{ psi}$$

$$(\sigma_m) = 363.9 \text{ psi}$$

$$(n_y) = 2.1$$

4.6 Shaft Material Selection

For our shaft, we selected CD AISI 1045. For reasons similar to the gears, we chose this material because of its high strength, and given that this is for a small diameter, we did not want to cheap out to ensure the integrity of the structure. We almost selected AISI 1050 CD, because there was no improvement in diameter decrease and the factor of safety gain was negligible we ended up staying with AISI 1045 CD. Instead purchasing a more expensive material we found one that behaves just as good at a cheaper price which is significant to creating this geartrain.

5. Bearing Selection

5.1 Bearing Type

We selected roller bearings for the gear box as these can withstand large radial loads compared to ball bearings. These bearings are specifically designed to handle these types of loads over an extended period of time, making them much more reliable in long term use. This is exactly what we need for a gear box that is estimated to have an infinite life. Compared to ball bearings, the roller bearing selection decreases overall friction, thus improving the efficiency of the bearing as well as decreasing the likelihood of potential failures.

Utilizing this information, we calculated the C10 rating for the roller bearings based on the applied load from the shafts. This allowed us to approximate the standard bearing C10 rating, thus determining the appropriate bearing sizes that can be utilized within the gear box.

5.2 Bearing Load Analysis

From previous analysis, we can use the following reactions to support bearing selection:

$$R_1 = 67.07 \text{ lbf}$$

$$R_p = 109.83 \text{ lbf}$$

$$R_G = 259.48 \text{ lbf}$$

$$R_3 = 223.57 \text{ lbf}$$

5.3 Bearing Life Calculation

$$\text{Manufacturer 2: } L_R = 1(10)^6 \quad X_0 = 0.02 \quad \theta = 4.459 \quad b = 1.483$$

$$a_f = 1.2 \quad R_D = 0.99 \text{ (Reliability of 99\%)} \quad a = \frac{10}{3} \text{ (Roller Bearings)} \quad L = 5000 \text{ hours}$$

$$X_D = \frac{L_D}{L_R} = \frac{L(n)(60)}{L_R}$$

$$\Rightarrow (X_D)_P = 600$$

$$\Rightarrow (X_D)_G = 54$$

$$C_{10} = a_f F_D \left[\frac{X_D}{X_0 + (\theta - X_0) \left[\ln \left(\frac{1}{R_D} \right) \right]^{\frac{1}{b}}} \right]^{\frac{1}{a}}$$

$$(C_{10})_1 = 864.4 \text{ lbf} \approx 3.85 \text{ kN}$$

$$(C_{10})_G = 986.4 \text{ lbf} \approx 4.39 \text{ kN}$$

$$(C_{10})_P = 2330.2 \text{ lbf} \approx 10.37 \text{ kN}$$

$$(C_{10})_3 = 1399.1 \text{ lbf} \approx 6.22 \text{ kN}$$

5.4 Bearing Selection Table

From the table provided, we are able to access the dimensions of the bearings to the corresponding load rating that was calculated.

Table 2. Load Rating Values

02-Series			02-Series			03-Series	03-Series	
Bore, mm	OD, mm	Width, mm	Load Rating, kN	Load Rating, kN	OD, mm	Width, mm	Load Rating, kN	Load Rating, kN
			C_{10}	C_0			C_{10}	C_0
25	52	15	16.8	8.8	62	17	28.6	15.0
30	62	16	22.4	12.0	72	19	36.9	20.0
35	72	17	31.9	17.6	80	21	44.6	27.1
40	80	18	41.8	24.0	90	23	56.1	32.5
45	85	19	44.0	25.5	100	25	72.1	45.4
50	90	20	45.7	27.5	110	27	88.0	52.0
55	100	21	56.1	34.0	120	29	102	67.2
60	110	22	64.4	43.1	130	31	123	76.5
65	120	23	76.5	51.2	140	33	138	85.0
70	125	24	79.2	51.2	150	35	151	102
75	130	25	93.1	63.2	160	37	183	125
80	140	26	106	69.4	170	39	190	125
85	150	28	119	78.3	180	41	212	149
90	160	30	142	100	190	43	242	160
95	170	32	165	112	200	45	264	189
100	180	34	183	125	215	47	303	220
110	200	38	229	167	240	50	391	304
120	215	40	260	183	260	55	457	340
130	230	40	270	193	280	58	539	408
140	250	42	319	240	300	62	682	454
150	270	45	446	260	320	65	781	502

6. Housing / Gearbox Case Design

6.1 Housing Requirements

For the housing, there are several requirements it must follow. Since we want to make the gear train as cost effective, we want it to be compact. We want to use the least amount of materials, get small manufactured parts to support small sizes and keep expenses to a minimum. Although we want the geartrain to be small, we want it to be durable. We are expected to give an infinite life per the shafts of the system. We would like for the gear box to have easy assembly and protect the gears along with it because if we do not, the gear train becomes inoperable. These are the main housing requirements we are to make true.

6.2 Material Selection

The material that we agreed to use on the housing is 6061 T6 Aluminum. We selected aluminum because its manufacturing costs are much lower than those of cast iron. This is due to it being much easier to machine. Additionally for the gear box that we are designing, it can support the environment. There is enough strength for bearing support, moderate gear forces, shaft displacement, and overall can remain rigid. Extreme stiffness is not required for our application. Additionally, aluminum is corrosion resistant. This is significant because we are advertising the tractor with capabilities of shoveling snow. If the gear train corrodes then that affects operation significantly. Overall, we feel that 6061 T6 Aluminum would work well as our gear trains housing.

6.3 Housing Dimensions

Housing dimensions are as follows (IPS):

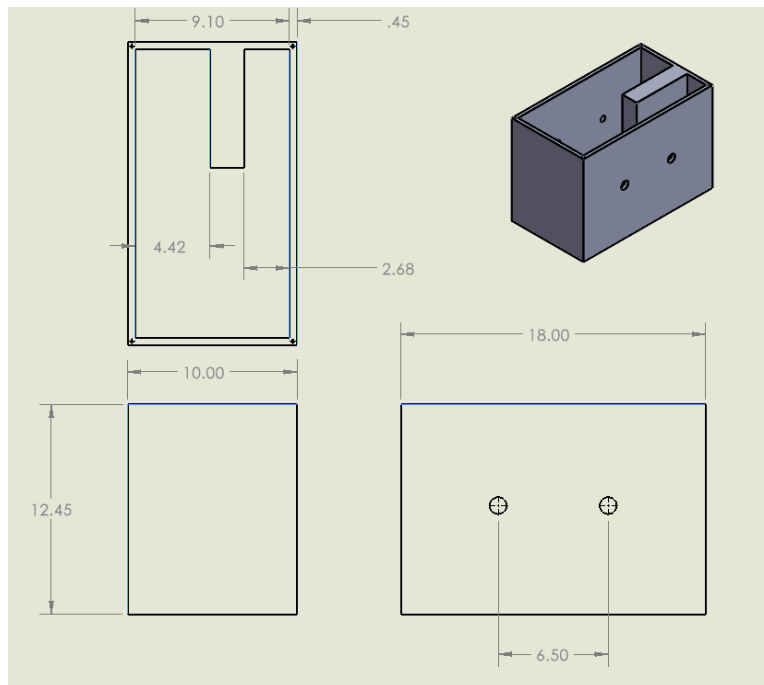


Figure 10. SolidWorks Case Drawing

6.4 Stress Considerations

When considering the stress on housing, there are many areas that can be considered. For starters, the bearing seat stresses are localized and compressive. Ridged housing must be ensured so that shafts properly align and gears mesh as desired. Along with the bearing stress is the wall stresses. The wall can potentially endure bending, flexing etc. Since the geartrain is receiving large amounts of revolutions per minute, we can expect vibrations and potential fluctuating torques, this is where cast iron proves itself to be better as its stiffness counter the vibrations but for what we're making, aluminum should be stable. We can also consider fatigue stresses, even if gears operate under yielding stress, repetitive loading can cause wear around holes, corners, or bolts. The final consideration would be temperature. Heat is generated during operation which can affect components and cause stresses to increase.

7. Screw / Fastener Design

7.1 Load Analysis

From earlier analysis, we know T_{in} and T_{out} .

$$T_{house} = T_{in} + T_{out} = 189.07 + 2100.85 = 2289.93 \text{ lbf in}$$

$$D_{Screw(Corner)} = 8.20"$$

$$D_{Screw(Left/Right)} = 7.0"$$

$$D_{Screw(Top/Bottom)} = 5.0"$$

$$\Sigma r^2 = 4(8.2")^2 + 2(7")^2 + 2(5")^2 = 416.96 \quad F_{Screw} = \frac{T(D)}{\Sigma r^2}$$

$$F_{Screw(Corner)} = 45.03 \text{ lbf}$$

$$F_{Screw(Left/Right)} = 38.44 \text{ lbf}$$

$$F_{Screw(Top/Bottom)} = 27.46 \text{ lbf}$$

7.2 Screw Discussion & Selection

Since the fasteners being used are not power screws, analysis upon stresses is unfamiliar.

However, we have calculated the forces amongst each of our 8 fasteners and found the largest load that must be endured is 45.0 lbf. Knowing that, after searching we found that

$M5 \times 0.8$ or #10 – 32 UNF have capabilities of enduring well over 45 lbf. To simplify our design, we are selecting 1 fastener to support the housing. Although “over engineered”, the fasteners are relatively cheap and will not affect the cost as other components do. We potentially would like to opt for UNF as they handle vibrations greater than a course thread.

Table 2. Screw Parameters

Property	10-32 UNF
Diameter	0.19"
Thread Pitch	32 $\frac{Threads}{in}$
Head Size	0.3125 in
Shear Capacity	550 lbf

8. CAD Modeling and Assembly

8.1 Individual Part Models

Gears

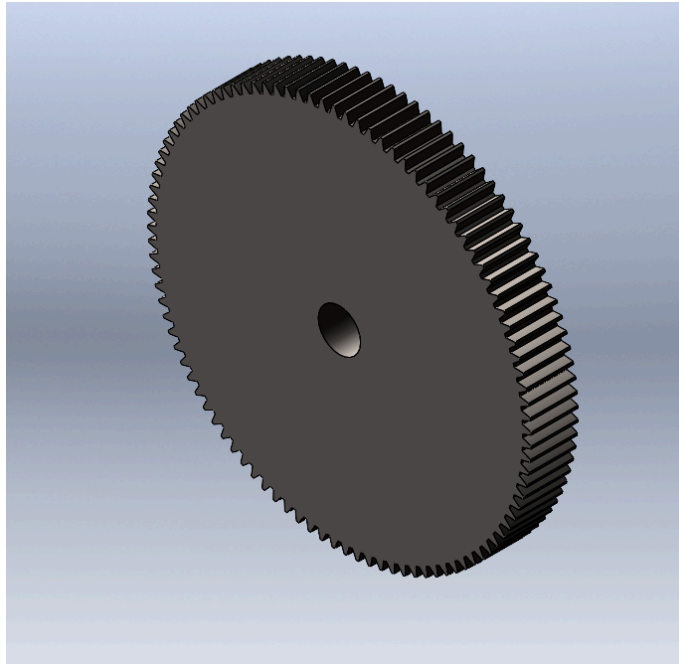


Figure 11. Gear 100 Teeth

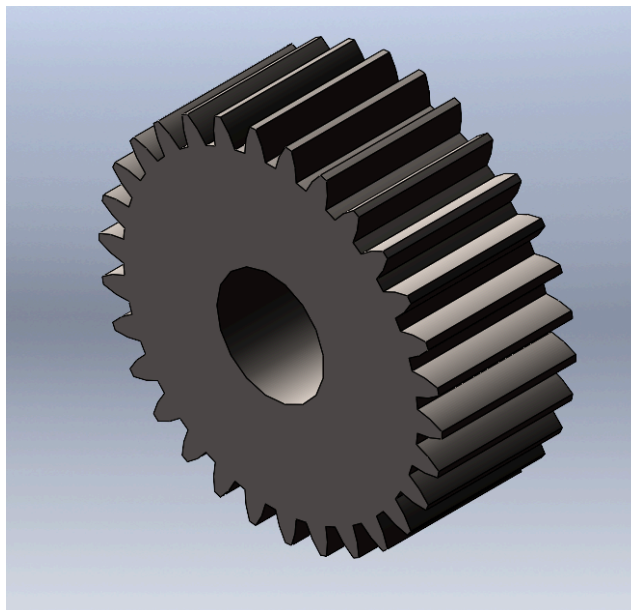


Figure 12. Pinion 30 teeth

Shafts

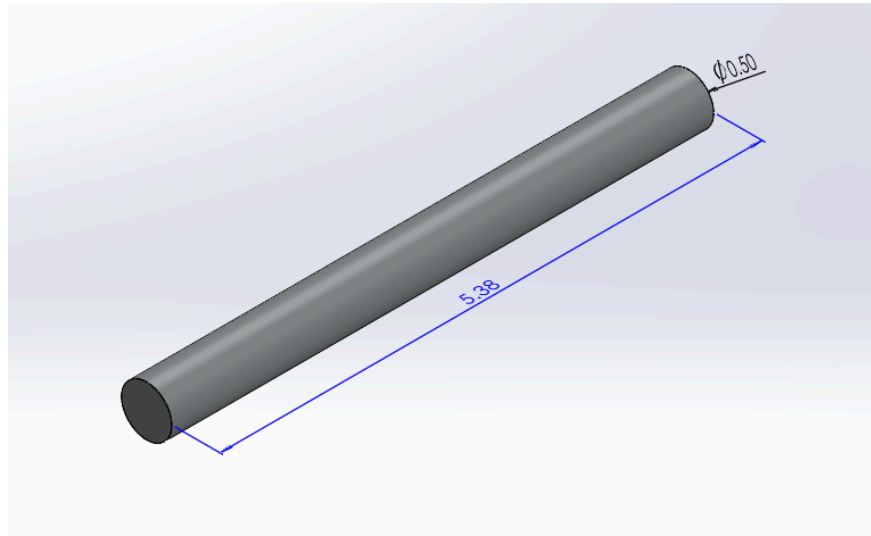


Figure 13. Input Shaft

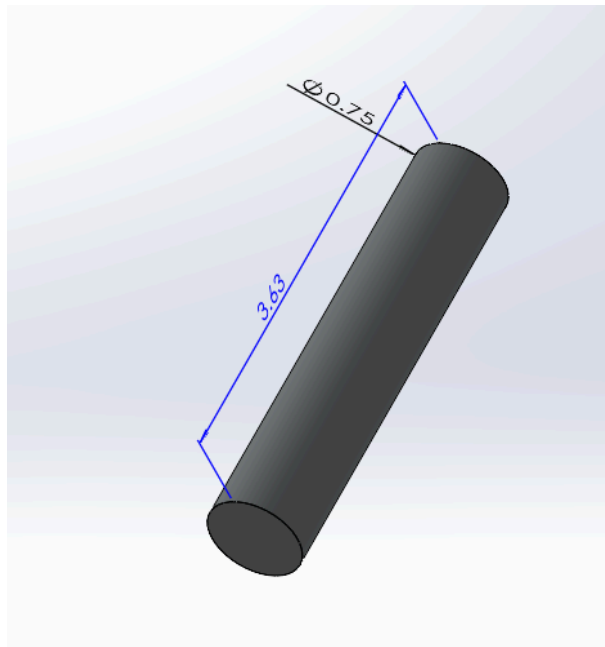


Figure 14. Output Shaft

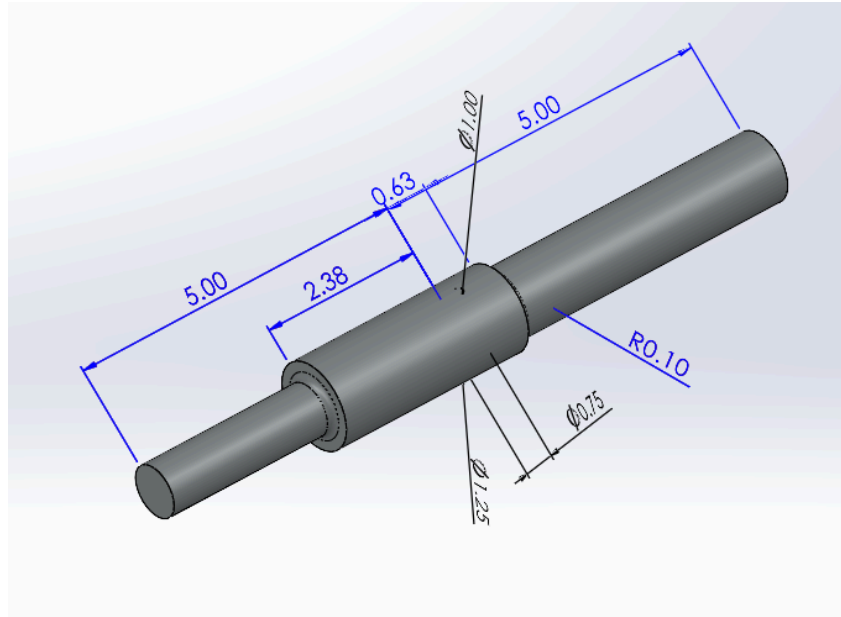


Figure 15. Stepshaft

Housing

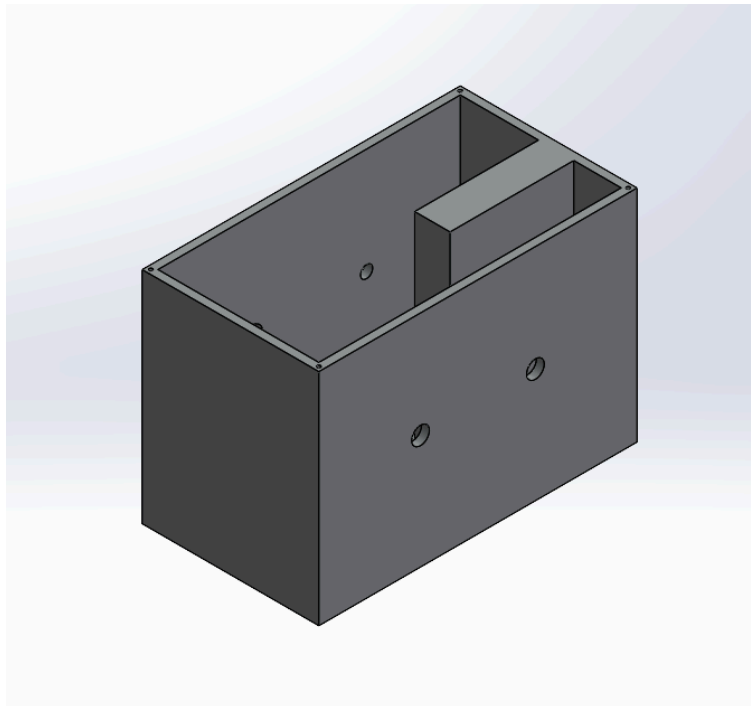


Figure 16. Housing

Bearings

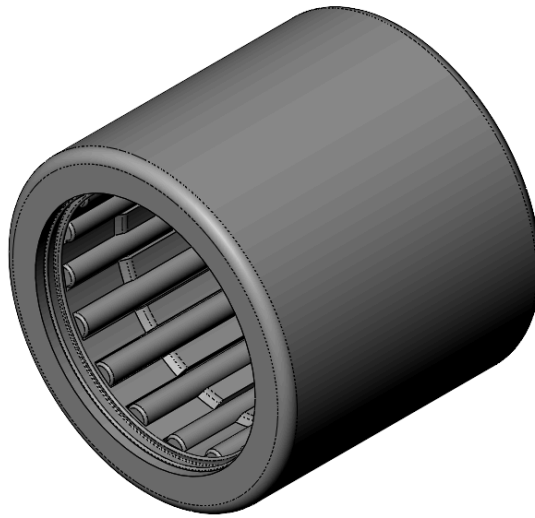


Figure 17. 0.5ID_Needle-Roller Bearing

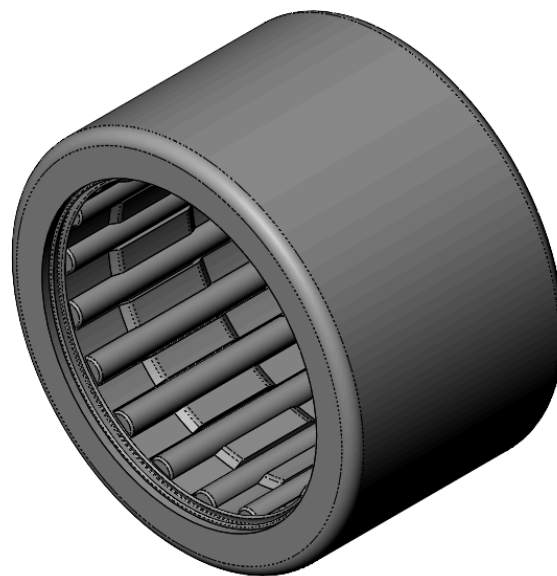


Figure 18. 0.75ID_Needle-Roller Bearing

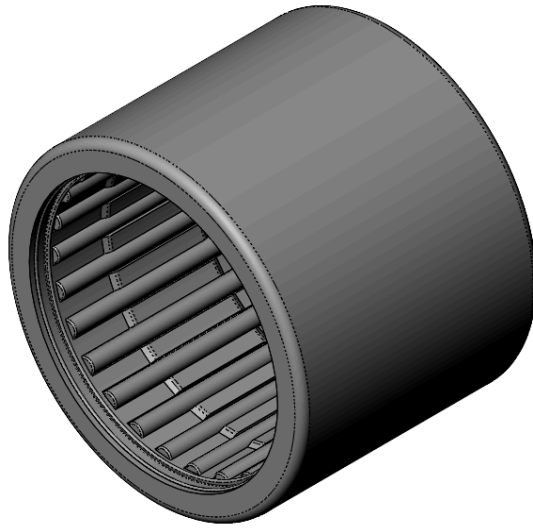


Figure 19. 1.0ID_Needle-Roller Bearing

8.2 Full Assembly

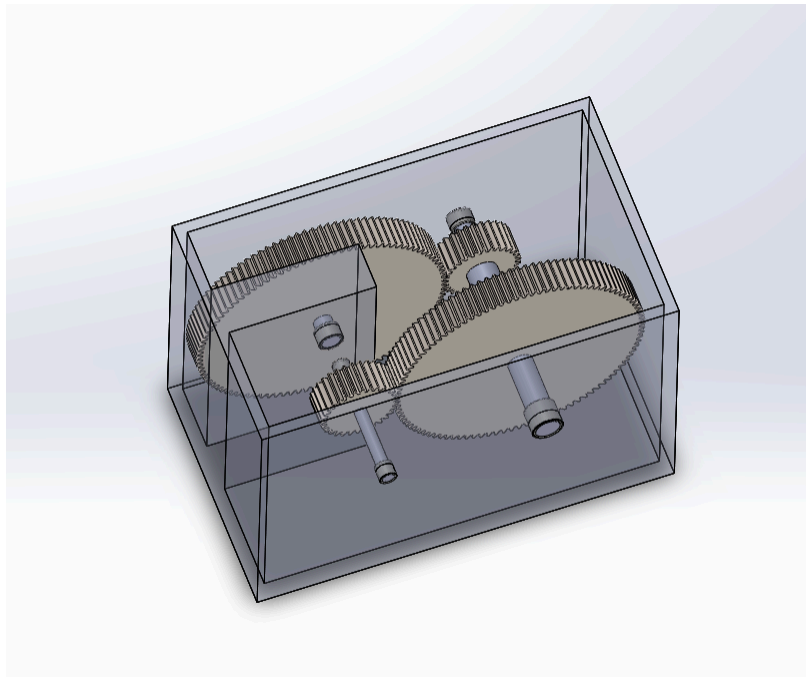


Figure 20. Isometric View of full assembly

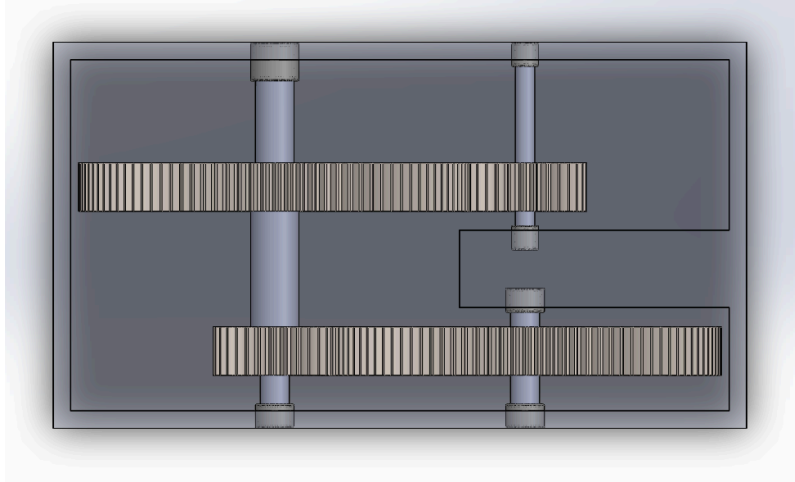


Figure 21. Top View of full assembly

9. SolidWorks FEA Analysis

9.1 Gear FEA

The FEA conducted on the spur gear in SolidWorks was used to study the stress distribution developed on the gear teeth during operation. The simulation focused on the bending and contact stresses generated as torque was transmitted through the gear drive. From the Von Mises stress distribution plot, the maximum stress on the gear was approximately 31,723 psi, while lower stress regions were distributed throughout the rest of the gear body. The highest stress concentrations were located near the gear tooth contact regions and at the tooth roots. This was expected since the gear teeth experience the greatest loading during torque transmission between mating gears, resulting in bending and compressive stresses in those areas.

The majority of the gear body experienced relatively low stress values, indicating that the gear operated under stable loading conditions. In addition, the maximum Von Mises stress value was well below the yield strength of approximately 76,700 psi for AISI 1045 steel. This indicates that the gear would operate safely without undergoing plastic deformation under the applied loads. Even though the stress values obtained were realistic, some stress concentrations may have been influenced by mesh refinement near the tooth edges and by assumptions made in the contact conditions. However, the overall stress distribution pattern matched the expected mechanical behavior of the gear during load transmission. Therefore, the SolidWorks FEA results helped validate the gear stress calculations obtained from MATLAB.

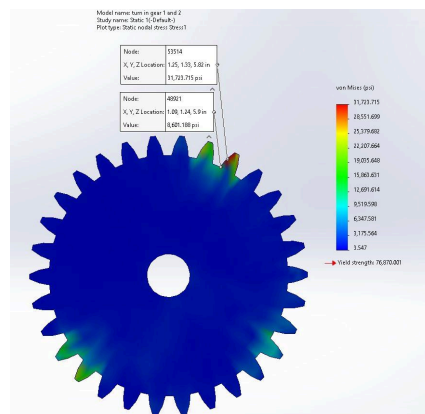


Figure 22. Gear Stress

9.2 Shaft FEA

Stress analysis of the shaft assembly was conducted in SolidWorks using the finite element method. The results from the Von Mises stress analysis showed that the maximum stress induced in the shaft was approximately 36,024 psi. The highest stresses were observed near the shaft mounting and transition regions. These areas experienced the greatest stress concentrations because they were subjected to torque transmission, support reactions, and localized stresses caused by geometry changes. Most regions of the shaft experienced relatively low stress levels, indicating that the shaft was capable of distributing the applied loads effectively throughout its structure. The most highly stressed regions were associated with the shaft mounting and support features, which was expected since these areas experience higher loads during operation.

From the simulation results, it was observed that the maximum Von Mises stress was lower than the yield strength of the selected AISI 1045 steel material. This indicates that the shaft would operate safely under the specified loading conditions without experiencing permanent deformation. In addition, the stress simulation results were consistent with the shaft analysis performed using the MATLAB Shaft tool, which predicted acceptable factors of safety for the gearbox shafts. Although some local stress concentrations may have been influenced by the mesh density used in the simulation, the overall stress behavior of the shaft was consistent with the expected mechanical response of a shaft subjected to transmitted torque and support reaction forces.

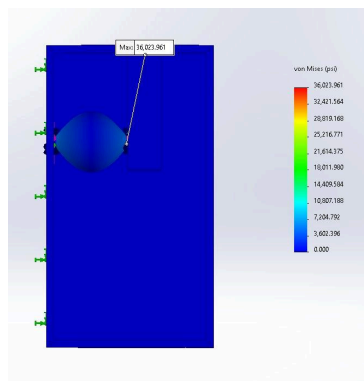


Figure 23. Stresses

10. Second FEA Software Analysis

The second FEA software analysis will be conducted using Autodesk Inventor to confirm the findings from the SolidWorks simulation test. This will be accomplished by importing the gearbox assembly designed in SolidWorks into Autodesk Inventor for static stress analysis. The static stress analysis will involve applying material properties, boundary conditions, loads, and meshing to the different components such as the gears, shafts, and housing within the Autodesk Inventor environment. The results from the Autodesk Inventor stress analysis will then be compared to the SolidWorks simulation results for accuracy and mesh sensitivity.

10.1 Software Setup

Mesh

To create a mesh model of the gearbox assembly in Autodesk Inventor, the assembly was divided into smaller finite elements that could be analyzed in terms of stress. The initial simulation used a medium mesh density to achieve accurate results within a reasonable amount of simulation time. In addition, automatic mesh refinement was applied near the gear teeth, shaft fillets, and regions where contact loads were expected in order to account for possible stress concentration areas. The mesh allowed the analysis of stress distribution, deformation, and factor of safety throughout the gearbox assembly.

Boundary Conditions

The boundary conditions were set up to simulate realistic operating conditions of the gearbox. To make the simulation as realistic as possible, a fixed constraint was applied to the bottom surface of the gearbox assembly to represent the gearbox being mounted to the tractor frame. Next, a moment load was applied to the input shaft to represent the torque transmitted from the tractor's engine through the gearbox and gear train. Automatic bonded contacts were used to simulate the contact interactions between the various components.

Properties of Materials

Material properties were assigned to each of the components that made up the gearbox based on the selected design materials. For the gearbox housing and cover, Aluminum 6061 was selected

due to its light weight, corrosion resistance, and sufficient rigidity for this application. For the gears and shafts, Steel AISI 1045 HV HR was selected because of its high strength, wear resistance, and cost effectiveness for power transmission applications.

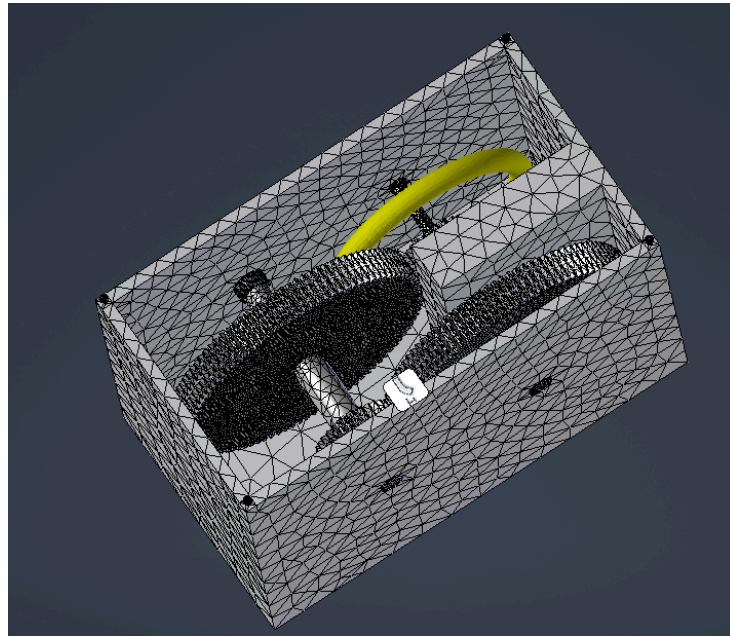


Figure 24. Mesh Analysis View on Inventor

10.2 Results

Stress Results

In the stress analysis performed using Autodesk Inventor, the input shaft had a maximum Von Mises stress of about 48.73 ksi under the specified moment loading condition. There was low stress across most of the gearbox assembly, but higher stresses were found in the shaft transition areas and shaft to gear joint regions. The high stress concentration in these regions is expected since the shaft is subjected to bending and torsional loads while transferring torque through the gears. The stress results obtained for the chosen AISI 1045 steel material are reasonable and are consistent with the MATLAB results.

In the first principal stress analysis performed on the shaft assembly, the highest tensile stress was 27.11 ksi. Tensile stresses developed in the shoulder and fillet areas where there are

geometric discontinuities. This is normal for rotating shafts since they contain regions where the diameters change abruptly.

In the third principal stress analysis performed on the shaft assembly, the lowest compressive stress was -31.63 ksi. Compressive stress resulted from the loads and reaction forces applied to the shaft during torque transfer. The compressive stress distribution throughout the gearbox assembly was fairly even except in the shaft regions.

Even though the results seem physically reasonable, some errors may still exist due to assumptions made during the finite element modeling process. Automatic bonded contact was applied between the components, which may have caused overconstraint in certain regions and resulted in stress concentrations that are slightly higher than what would occur in real-world conditions. In addition, the STEP file geometry, mesh density near sharp corners, and assumptions made for the boundary conditions may have also contributed to the stress peaks observed in the model.

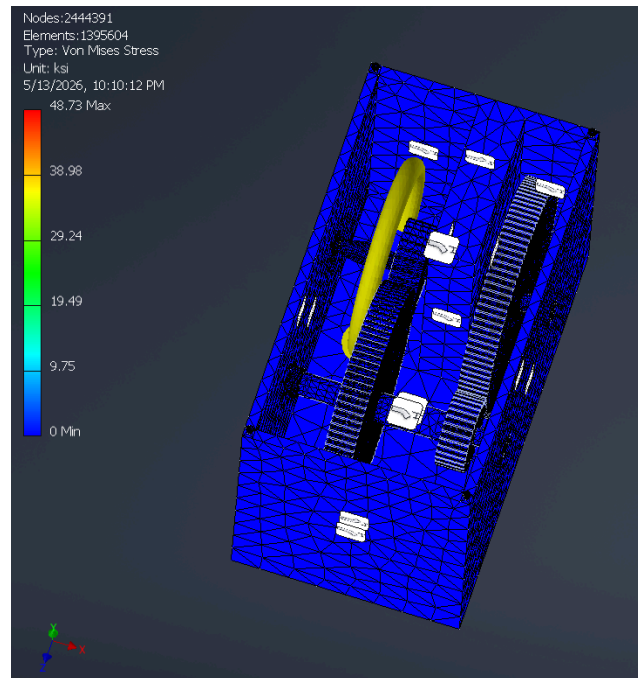


Figure 25. Input Shaft Von Mises Analysis on Inventor

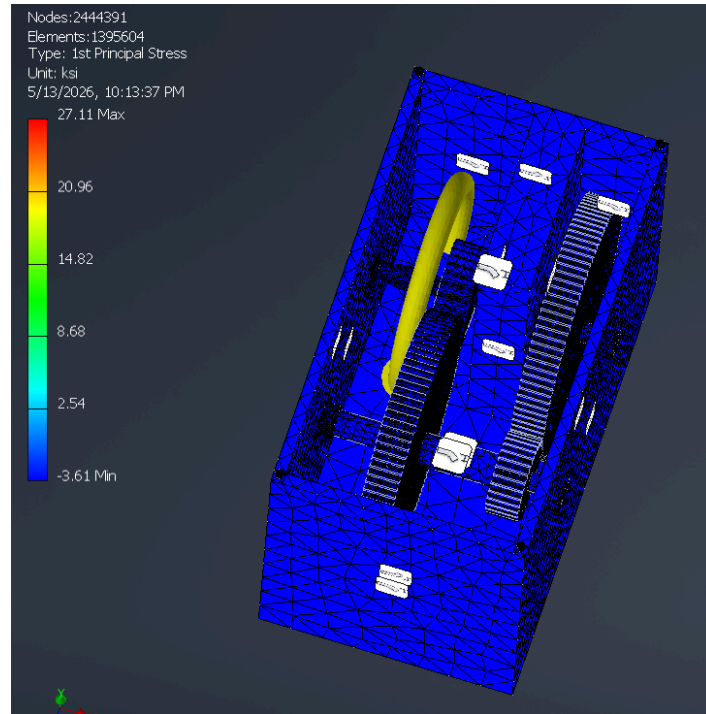


Figure 26. Input Shaft 1st Principal Stress Analysis on Inventor

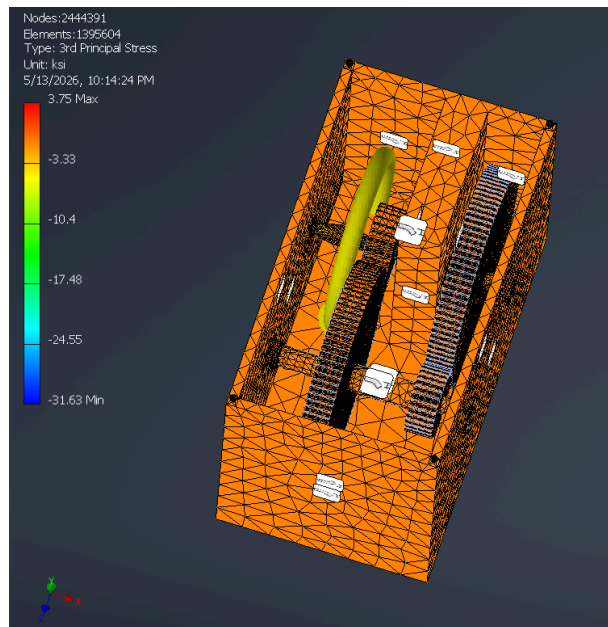


Figure 27. Input Shaft 3rd Principal Stress Analysis on Inventor

Deformation Results

The displacement analysis results obtained using Autodesk Inventor indicate that the maximum deformation was approximately 5.208×10^{-4} inches under the applied loading conditions. These deformations were mainly found in the regions surrounding the loaded gears and shafts where torque transfer occurs. Overall, the deformation throughout the gearbox was very low, indicating that the gearbox casing and shaft structure had sufficient stiffness to resist deformation under the applied loads.

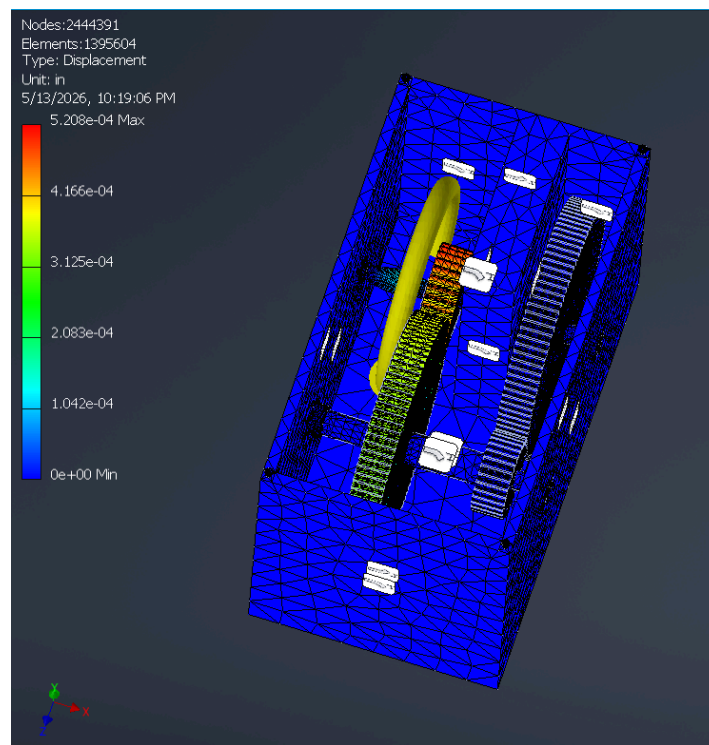


Figure 28. Deformation results from Inventor

Results for Factor of Safety

For the factor of safety analysis, Autodesk Inventor predicted a minimum factor of safety of approximately 0.74 in localized regions of the gearbox assembly. The lowest factor of safety values occurred near the shaft transitions and gear mounting areas where the highest stress values were observed. Most regions of the gearbox assembly showed a factor of safety greater than one,

indicating that the majority of the assembly experienced low stress concentrations during the simulation.

Even though some localized regions had a factor of safety lower than 1, these results are likely due to artificial stress concentrations caused by assumptions used in the finite element analysis. For example, automatic bonded contacts, idealized boundary conditions, sharp geometry transitions, and mesh refinement near certain areas can produce localized stress peaks in the model. In comparison, the MATLAB analytical analysis predicted shaft factors of safety greater than 1.5. Therefore, the shaft design appears to have an acceptable factor of safety under the applied loading conditions of the gearbox assembly. Overall, the gearbox assembly demonstrated reasonable behavior in terms of deformation and stress distribution.

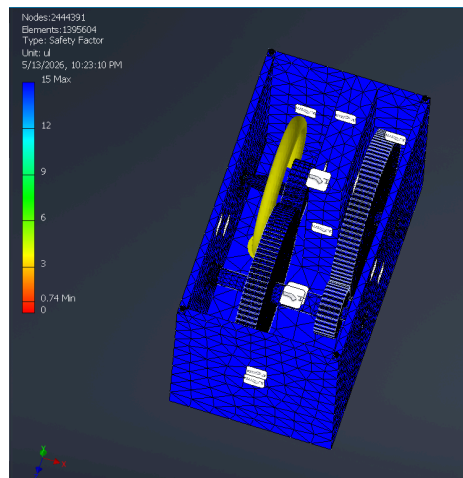


Figure 29. Factor of Safety from Inventor

10.3 Comparison with SolidWorks

Table 3. Stress comparison

Component	Theoretical Stress (psi)	Simulation Stress (psi)	Percent Offset
Gear (Bending)	6295.74	8601.188	36.62%
Gear (Contact)	54035.88	31723.715	41.29%
Shaft	17248.76	36023.961	108.85%

11. MATLAB Calculations

11.1 Gear Calculations

To calculate the number of teeth we first had to find the gear ratio we were looking for which was a total gear ratio of $2000/180=11.11$. Since the train value was larger than 10 we had to do this in 2 stages so to use the same gear sets we square rooted this to get 3.33 for our goal gear ratio. Next we had to find the minimum number of teeth using the formula

$$N_p = \frac{2k}{(1+2m)\sin(\phi)^2} (m + \sqrt{m^2 + (1 + 2m)\sin(\phi)^2}).$$
 This gave us a minimum of 15.16 which

we rounded up to 16. Then keeping the as close to the same tooth gear ratio we got the number of teeth on the gear to be 53. These teeth dimensions caused some issues due to availability and also the bore diameter being far too large for the pinion on the countershaft. This led to us finding another gearset that would keep a similar gear ratio and led to us choosing a pinion with 30 teeth and a gear with 100 teeth at a diametral pitch of 10 teeth/inch. This helped in keeping the bore diameter from being too large relative to the gears and possibly weakening the gears.

11.2 Shaft Calculations

Distortion Energy Goodman criteria was used for testing for failure on the shafts. We decided to use a factor of safety of 1.5 for the minimum on the shafts. The input and output shafts were simple to set up as they had no diameter changes and the only spot to check was at the peak moment on those. I assumed fully sensitive at first and did 3 iterations before settling on a value for those. For the counter shaft I had to check three locations. Those were the first fillet after the gear, the second fillet before the pinion, and at the pinion. This was to find a minimum diameter for these as well as for checking the diameter change that was acceptable for torsion and bending stress. This gave us a diameter of $\frac{1}{2}$ inches for the input shaft, $\frac{3}{4}$ inches for the output shaft, and the countershaft has a diameter of $\frac{3}{4}$ inches for the gear, $1 \frac{1}{4}$ inches for the step, and a diameter of 1 inch for the pinion.

11.3 Verification of Results for Gears and Shaft

```
%calculate at root near gear 4 since it is has a stress concentration

ke=(1-0.08*2.326);          % Reliability Marin Factor
ka=2*(Sut)^-0.217;          % Surface Marin Factor
kb=0.9;                      % guess of kb value
Se=Seg*ka*kb*ke;            % initial Se with guesses

% Calculating at root before the pinion
A=sqrt(4*(Kt*Mroot)^2);      % Alternating Stress
B=sqrt(3*(Kts*Tm2)^2);       % Mean Stress
d=((((16*FOS)/pi)*((A/(Se*10^3)))+(B/(Sut*10^3))))^1/3; % Starting point for Diameter

% Calculating at gear since it has the largest moment applied

A=sqrt(4*(Mtot2)^2);         % Alternating Stress
B=sqrt(3*(Tm2)^2);          % Mean Stress
dg=((((16*FOS)/pi)*((A/(Se*10^3)))+(B/(Sut*10^3))))^1/3;
r=d/10;

% Calculating at root of first Gear for min diameter

A1=sqrt(4*(Kt1*Mroot1)^2);
B1=sqrt(3*(Kts1*Tm2)^2);
d1=((((16*FOS)/pi)*((A1/(Se*10^3)))+(B1/(Sut*10^3))))^1/3;
r1=d1/10;
```

```
343 % Final FOS check for diameter at second root
344 A=sqrt(4*(Kf*Mroot)^2);
345 B=sqrt(3*(Kfs*Tm2)^2);
346 n=((((pi*d^3)/16)*((A/(Sen*10^3)))+(B/(Sut*10^3))))^-1);
347 |
348 % Final FOS check for diameter at first root
349 A1=sqrt(4*(Kf1*Mroot1)^2);
350 B1=sqrt(3*(Kfs1*Tm2)^2);
351 n1=((((pi*d1^3)/16)*((A1/(Sen1*10^3)))+(B1/(Sut*10^3))))^-1);
352 |
353 % Final check for for diameter at gear
354 dg=ceil(dg*8)/8;
355 kbg=(dg/0.3)^-0.107;
356 Seng=Seg*ka*kb*ke;
357 A3=sqrt(4*(Mtot2)^2);
358 B3=sqrt(3*(Tm2)^2);
359 ng=((((pi*dg^3)/16)*((A3/(Seng*10^3)))+(B3/(Sut*10^3))))^-1);
360 |
361 n=min(n,ng);
362 n=min(n,n1);
363 % Check for yielding
364 Ny2=(Sy*10^3)/((((32*Kf*Mroot)/(pi*d^3))^2)+(3*((16*Kfs*Tm2)/(pi*d^3))^2))^(1/2));
365 Ng=(Sy*10^3)/((((32*Mtot2)/(pi*dg^3))^2)+(3*((16*Tm2)/(pi*dg^3))^2))^(1/2));
366 Ny1=(Sy*10^3)/((((32*Kf1*Mroot1)/(pi*d1^3))^2)+(3*((16*Kfs1*Tm2)/(pi*d1^3))^2))^(1/2));
```

Command Window

```
Calculate FOS for our 2nd gear see 13.1.1.2.70
The bore diameter for the first gear on the countershaft is 0.75 inches
The bore diameter for the second gear on the Counter shaft is: 1 inches
The large diameter of Counter shaft is: 1.25 inches
The radius of fillet is: 0.1 inches
The Factor of Saftey of Counter shaft is: 1.5
Yeilding Factor of Safetyv of the Counter shaft is: 3.5837
```

11.4 Bearing Calculations

The specific C10 formula utilized in this process can be seen within the MATLAB code along with a few parameters previously mentioned. The load-life relationship for a bearing utilizes the

constant, a , for roller bearings as $10/3$. We then used an application factor of 1.2, the same application factor used in several industry applications. Using the Weibull Parameters for bearings, we used Manufacturer 2 parameters allowing us to use a rating life of 10^6 , x sub-zero as 0.02, θ as 4.459, and b as 1.483. These parameters are utilized for ball and straight roller bearings. Lastly, we are using a reliability of 99% which will increase the life expectancy of the bearings themselves. The load life of the individual bearings were calculated based on its operation hours, 5000, and angular speed. The multiple of rating life, was then computed based on the load life of the bearing divided by the rating life of Manufacturer 2 parameters, 10^6 .

Utilizing the resultant forces from the shafts being applied to the bearings, the C10 rating life for the bearings on shaft 1 is 864.4 lbf or 3.85 kN. The C10 rating life for the bearings near the input and near the output on shaft 2 are 986.4 lbf or 4.39 kN and 2330.2 lbf or 10.37 kN, respectively. Lastly the C10 rating life for the bearings on shaft 3 are 1399.1 lbf or 6.22 kN.

11.5 Verification of Results for Bearings

```
% Bearing Info
AppF=1.2;
Rel=0.99;
a=3/10;
Ophr=5000;
X0=0.02;
Theta=4.459;
b=1.483;
Lr=10^6;
Bot=X0+(Theta-X0)*(log(1/Rel))^(1/b));

% Shaft 1

R1=sqrt(Raz1^2+Ray1^2);
Plrpm=2000;
LDP1=Plrpm*Ophr*60;
XDP1=LDP1/Lr;
C10P1=(AppF*R1)*(XDP1/Bot)^(a);

% Countershaft

Ra2=sqrt(Raxz^2+Raxy^2);
Glrpm=Plrpm*30/100;
LDG1=Glrpm*Ophr*60;
XDG1=LDG1/Lr;
C10G1=(AppF*Ra2)*(XDG1/Bot)^(a);

Rb2=sqrt(Rbxz^2+Rbxy^2);
P2rpm=Glrpm;
LDP2=P2rpm*Ophr*60;
XDP2=LDP2/Lr;
C10P2=(AppF*Rb2)*(XDP2/Bot)^(a);

% Shaft 3

R3=sqrt(Raz2^2+Ray2^2);
G2rpm=Glrpm*30/100;
LDG2=G2rpm*Ophr*60;
XDG2=LDG2/Lr;
C10G2=(AppF*R3)*(XDG2/Bot)^(a);

disp(['The bearings for shaft 1 need a C10 of : ', num2str(ceil(C10P1*10)/
10), ' lbf'])
disp(['The bearing near the Input gear for shaft 2 needs a C10 of : ',
num2str(ceil(C10G1*10)/10), ' lbf'])
disp(['The bearing far from the Input gear for shaft 2 needs a C10 of : ',
num2str(ceil(C10P2*10)/10), ' lbf'])
disp(['The bearings for shaft 3 need a C10 of : ', num2str(ceil(C10G2*10)/
10), ' lbf'])

The bearings for shaft 1 need a C10 of : 864.4 lbf
The bearing near the Input gear for shaft 2 needs a C10 of : 986.4 lbf
The bearing far from the Input gear for shaft 2 needs a C10 of : 2330.2 lbf
The bearings for shaft 3 need a C10 of : 1399.1 lbf
```

11.6 Screw Calculations

To compute the total shear force that the screws would undergo, we utilized the distance between the center of the plate to the outer corners and sides of the plate. Using the total torque that the housing would undergo from both the input and output combined, we were able to find the shear

forces on each screw. The resultant values for the corner screws, the left and right side screws, and top and bottom screws were 38.7012 lbs, 33.3579 lbs, and 19.6223 lbs, respectively.

11.7 Verification of Results for Screws

```
% Finding force applied onto each screw

Tin=Ft23*dp/2;
Tout=Ft45*Dg/2;
Thouse=Tout+Tin;

% Distances

L=Dbc;
H=(Dg/2)+(dp/2)+2*((Dg/2)+(1/P))+0.3;
DTS1=sqrt((L/2)^2+(H/2)^2);
DTS2=H/2;
DTS3=L/2;
DTtot=(4*(DTS1^2)+2*(DTS2^2)+2*(DTS3^2));

%Torque on each screw
TPS1=Thouse*DTS1/DTtot;
TPS2=Thouse*DTS2/DTtot;
TPS3=Thouse*DTS3/DTtot;

disp(['The 4 screws on the corners must be able to resist ', num2str(TPS1),
' lbs of shear force'])
disp(['The left and right screws must be able to resist ', num2str(TPS2), '
lbs of shear force'])
disp(['The top and bottom screws must be able to resist ', num2str(TPS3), '
lbs of shear force'])

The 4 screws on the corners must be able to resist 38.7012 lbs of shear force
The left and right screws must be able to resist 33.3579 lbs of shear force
The top and bottom screws must be able to resist 19.6223 lbs of shear force
```

12. Cost Analysis

12.1 Total Gearbox Cost

Table 4. Cost breakdown

Part	Quantity	Individual Price (\$)	Total Price (\$)
Pinion (Shaft 1)	1	210.14	210.14
Gear (Shaft 2)	1	613.64	613.64
Pinion (Shaft 2)	1	210.14	210.14
Gear (Shaft 3)	1	613.64	613.64
Shaft 1	4 inches	0.64/inch	2.56
Shaft 2	10 inches	1.46/inch	14.60
Shaft 3	4 inches	1.00/inch	4.00
Bearing (Shaft 1)	1	16.71	16.71
Bearing (Shaft 2)	2	12.87	25.74
Bearing (Shaft 2)	2	12.14	24.28
Bearing (Shaft 3)	1	12.87	12.87
Fasteners	8	.176	1.408

13. Discussion

13.1 Comparison of MATLAB and FEA Results

The MATLAB analytical calculations and Autodesk Inventor FEA analyses produced similar trends in the structural behavior of the gearbox components. Both analyses showed that stress concentrations caused by torque application and geometric transitions led to maximum stresses near shaft transitions, fillet areas, and gear mounting regions. The maximum Von Mises stress obtained from the Inventor analysis was about 48.73 ksi for the input shaft. The stresses calculated using MATLAB were also within a reasonable operating range for AISI 1045 steel.

Very small deformation values were observed for all gearbox components in the Inventor simulation. In particular, the maximum displacement was 5.208×10^{-4} inches. This result supported the MATLAB findings and confirmed that the gearbox was rigid enough to withstand the expected forces and torques.

The MATLAB calculations predicted factors of safety greater than 1.5 for all gearbox shafts. In contrast, the FEA simulation produced some localized regions with factors of safety lower than 1. However, these regions were likely caused by stress singularities, bonded contact assumptions, mesh sensitivity, and simplified boundary conditions used in the finite element analysis.

13.2 Design Strengths

Another advantage of the gearbox design is its small size. Thanks to the usage of a double reduction spur gear, the gearbox managed to have a good speed reduction ratio with relatively small gears and an overall small size of the assembly. Also, materials used in the construction are strong enough to withstand the loads and at the same time cheap. Steel AISI 1045 was chosen as the material for gears and shafts because it is strong, not easily affected by wear, easy to machine and inexpensive. Also, aluminum 6061 was chosen for the housing since it is light-weighted, corrosion resistant and easy to manufacture.

In addition, the design proved to have relatively high factors of safety during calculations but at the same time showed low deformations of all the parts in the assembly. Moreover, the gearbox parts were manufactured by common manufacturing processes and consisted of off-the-shelf parts like roller bearings and bolts.

13.3 Design Limitations

While the gearbox system met all the project requirements, there are still some limitations associated with the design. First, there is the possibility of vibration and noise generation due to the use of spur gears. Spur gears tend to produce more noise because of the abrupt engagement of the gear teeth during rotation. Second, the assembly and alignment of multiple shafts, gears, and bearing systems can be difficult. Improper alignment may increase wear and reduce the overall lifespan of the components. Lastly, even though the gearbox housing was relatively compact, the use of steel gears and shafts increased the overall weight of the system. Over time, repeated loading and vibration could also lead to wear in high stress regions.

13.4 Possible Improvements

There are several improvements that could be made to further enhance the gearbox design. One improvement would be replacing the spur gears with helical gears to reduce vibration and noise while providing smoother power transmission since they have a line to point contact instead of just line contact. The gearbox could also benefit from different combinations of gears to potentially reduce the size of it as the 10"x17"x17" would be quite large for certain applications.

Another possible improvement would be making a lubrication system within the gearbox casing to promote smoother operation and reduce wear on the gear teeth. Additional improvements could include changes to the gearbox casing to reduce weight, as well as implementing more advanced contact modeling methods in the finite element analysis. If the team utilized various other types of analysis for the simulations by utilizing other softwares, the overall resultant values could be much more accurate. This allows the team and other engineers to determine the quality and accuracy of the data.

14. Conclusion

Our goal was to design a speed reducer gearbox that would survive a high number of cycles accounting for various factors. This is to be used in a commercial tractor and was to be at minimum two stages since the gear train value was larger than 10. In order to design the gear box

we had to calculate the forces acting on the gears, shafts, housing, and fasteners. From our results all components are able to survive the 5000 operation hours and we were able to achieve the output revolutions per minute of 180. Our design was using a compound reverted gear train so that we could minimize the space we took up and to minimize the number of skews and parts used in this so as to make it easier to maintain..

Similarly, the gearbox structure demonstrated acceptable performances under the loading conditions considered. The MATLAB predictions indicated that the shaft factors of safety were greater than 1.5, while the stress distribution and deformation results obtained from the FEA analysis were reasonable and showed small amounts of deformation throughout the gearbox assembly. Although stress concentrations were observed in certain areas of the gearbox structure due to assumptions and small mesh sizes, the reactions in those areas remained consistent with the mechanical behavior of the gearbox. The selected materials were also appropriate for the application, since AISI 1045 steel was used for the gears and shafts, while Aluminum 6061 was used for the housing. These materials provided a good balance between strength, durability, and weight.

One of the major advantages of the current design is its compactness and manufacturability. The components used, including the gears, bearings, and bolts, can be manufactured and assembled using various machining processes. In addition, the materials that were selected by the team offer good wear resistance and corrosion resistance, making the gearbox best for outdoor applications. However, spur gears generally have a higher chance to create vibration and noise compared to helical gears. In addition, the gearbox assembly contains many components where machining and alignment are critical to ensuring efficient system performance. Some inaccuracies were also observed in the finite element simulations due to contact assumptions, geometry conditions, and mesh sizes.

Overall, this project provided engineering experience in design, gear train analysis, shaft design, finite element analysis, CAD assembly modeling, and MATLAB programming. The project demonstrated that combining analytical calculations with FEA simulations can greatly improve an engineering design while also providing a deeper understanding of the mechanics involved.

Ultimately, this project helped strengthen our understanding of gearbox systems and mechanical power transmission.

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Appendices

Appendix A

MATLAB

```
clear
clc
format long g

%Gear Information

Ko=1;
Km=1.6;
Kb=1;
P=10;
Ks=1; % size variable
Cf=1;
mn=1; % 1 for spur gear can be calculated for spur gear
m=sqrt(2000/180); % gear ratio
k=1; % 1 for full depth
phi=20*pi/180; % angle converted to radians for calculations
np=2000;
N=60*5000*np;
F=1.25;
hp=6;
Q=6;
Yp=0.359;
Yg=.447;
Hb=250;

% Counter Shaft Information

% 1045 CD steel
Sut=91;
Sy=77;
Seg=Sut/2;

% Well rounded fillet assumption
Kt=1.62; % Assumed fully sensitive
Kts=1.35; % Assumed fully sensitive
Kt1=1.7;
Kts1=1.43;

% Shaft lengths
Dbc=10; % Total length of shaft
D1c=2; % distance from bearing 1 to first gear
D2c=4.25; % Distance between center of gears
D3c=3.75; % Distance from last gear to bearing 2
Drootc=D1c+D2c-F/2; % Distance from first bearing to fillet
FOS=1.5; % Factor of Safety

% First and Third shaft info
Db1=4; % Total length of shaft
D1=2; % Length from bearing to Gear
D2=2; % Distance from Gear to Bearing 2
```

```

% Gear Teeth Calculations
Np=((2*k)/((1+2*m)*sin(phi)^2))*(m+sqrt((m^2)+(1+2*m)*(sin(phi)^2)));

Np=ceil(Np);
Npnew=30;
Np=max(Npnew,Np);
disp(['Number of teeth on the pinion is: ', num2str(Np), ' Teeth'])

Ng=Np*m;

Ng=floor(Ng);
disp(['Number of teeth on the gear is: ', num2str(Ng), ' Teeth'])
mg=Ng/Np;

dp=Np/P;
Dg=Ng/P;
rp=dp/2;
rg=rp*mg;

C2C=(dp/2)+(Dg/2);
disp(['Diametral Pitch is: ', num2str(P), ' Teeth per inch'])

p=pi/P;

disp(['The circular pitch is: ', num2str(p), ' inchs'])

% First Gearset
v=pi*dp*np/12;
Wt=hp*33000/v;
J=0.39;

B=0.25*(12-Q)^(2/3);
A=50+56*(1-B);
Kv=((A+sqrt(v))/A)^B;

sigmaB=Wt*Ko*Kv*Ks*(P/F)*(Km*Kb/J);
disp(['Pinion Bending Stress is: ', num2str(sigmaB), ' PSI'])

St=77.3*Hb+12800;
Yn=1.3558*N^-0.0178;

Sf=St*Yn/sigmaB;
disp(['Bending FOS is: ', num2str(Sf)])

Cp=2100;
I=(cos(phi)*sin(phi)/(2*mn))*(mg/(mg+1));

sigmaC=Cp*sqrt(Wt*Ko*Kv*Ks*(Km/(dp*F))*(Cf/I));
disp(['Pinion Contact Stress is: ', num2str(sigmaC), ' PSI'])

Sc=322*Hb+29100;
Zn=1.4488*N^-0.023;

```

```

Sh=Sc*Zn/sigmaC;
disp(['Contact FOS is: ', num2str(Sh)])

% Second Gearset
v2=pi*dp*np/(12*mg);
Wt2=hp*33000/v2;
J=0.39;

B2=0.25*(12-Q)^(2/3);
A2=50+56*(1-B2);
Kv2=((A2+sqrt(v2))/A2)^B2;

sigmaB2=Wt2*Ko*Kv2*Ks*(P/F)*(Km*Kb/J);
disp(['Pinion Bending Stress for our 2nd gearset is: ', num2str(sigmaB2), '
PSI'])

N2=(5000*2000*60)/3.25;
St2=77.3*Hb+12800;
Yn2=1.3558*N2^-0.0178;

Sf2=St2*Yn2/sigmaB2;
disp(['Bending FOS for our 2nd gearset is: ', num2str(Sf2)])

I2=(cos(phi)*sin(phi)/(2*mn))*(mg/(mg+1));

sigmaC2=Cp*sqrt(Wt2*Ko*Kv2*Ks*(Km/(dp*F))*(Cf/I2));
disp(['Pinion Contact Stress for our 2nd gearset is: ', num2str(sigmaC2), '
PSI'])

Sc2=322*Hb+29100;
Zn2=1.4488*N2^-0.023;

Sh2=Sc2*Zn2/sigmaC2;
disp(['Contact FOS for our 2nd gearset is: ', num2str(Sh2)])

% First Gear forces
v1=pi*dp*np/12;
Ft23=33000*hp/v1;
F23=Ft23/cos(phi);
Fr23=F23*sin(phi);

%Second Gear forces
v2=pi*dp*np/(12*mg);
Ft45=33000*hp/v2;
F45=Ft45/cos(phi);
Fr45=F45*sin(phi);

% Solving for Z reactions, shears, and moments
Rbxz=(Ft23*D1c-Ft45*(D1c+D2c))/(Dbc);
Raxz=Ft23-Ft45-Rbxz;
Vxz=Raxz;
Vxz1=Vxz-Ft23;
Vxz2=Vxz1+Ft45;
Vxz3=Vxz2+Rbxz;

```

```

Mxz=Vxz*(D1c);
Mxz1=Mxz+Vxz1*(D2c);
Mxz2=Mxz1+Vxz2*(D3c);

% Solving for Y Reactions, Shears, and Moments
Rbxy=(-Fr23*D1c-Fr45*(D1c+D2c))/(Dbc);
Raxy=-Fr23-Fr45-Rbxy;
Vxy=Raxy;
Vxy1=Vxy+Fr23;
Vxy2=Vxy1+Fr45;
Vxy3=Vxy2+Rbxy;
Mxy=Raxy*(D1c);
Mxy1=Mxy+Vxy1*(D2c);
Mxy2=Mxy1+Vxy2*(D3c);

% Solving for Roots
Tm2=Ft23*Dg/2;
Mtot1=sqrt(Mxz^2+Mxy^2);
Mroot1xz=(Mxz+Vxz1*(F/2));
Mroot1xy=(Mxy+Vxy1*(F/2));
Mroot1=sqrt((Mroot1xz)^2+(Mroot1xy)^2);
Mrootxz=Mxz+Vxz1*(Drootc-D1c);
Mrootxy=Mxy+Vxy1*(Drootc-D1c);
Mtot2=sqrt(Mxz1^2+Mxy1^2);
Mroot=sqrt(Mrootxz^2+Mrootxy^2);

%calculate at root near gear 4 since it is has a stress concentration
ke=(1-0.08*2.326);
ka=2*(Sut)^-0.217;
kb=0.9; % guess of kb value
Se=Seg*ka*kb*ke; % initial Se with guesses

A=sqrt(4*(Kt*Mroot)^2);
B=sqrt(3*(Kts*Tm2)^2);
d=((16*FOS)/pi)*((A/(Se*10^3))+(B/(Sut*10^3)))^1/3;

% Calculating at gear since it has the largest moment applied
A=sqrt(4*(Mtot2)^2);
B=sqrt(3*(Tm2)^2);
dg=((16*FOS)/pi)*((A/(Se*10^3))+(B/(Sut*10^3)))^1/3;
r=d/10;

% Calculating at root of first Gear for min diameter
A1=sqrt(4*(Kt1*Mroot1)^2);
B1=sqrt(3*(Kts1*Tm2)^2);
d1=((16*FOS)/pi)*((A1/(Se*10^3))+(B1/(Sut*10^3)))^1/3;
r1=d1/10;

% Calculating neuber and sensitivity
neuber=0.246-3.08*(10^-3)*Sut+1.51*(10^-5)*Sut^2-2.67*(10^-8)*Sut^3;
neuberT=0.190-2.51*(10^-3)*Sut+1.35*(10^-5)*Sut^2-2.67*(10^-8)*Sut^3;
q=1/(1+(neuber/sqrt(r)));
qs=1/(1+(neuberT/sqrt(r)));
q1=1/(1+(neuber/sqrt(r1)));

```

```

qsl=1/(1+(neuberT/sqrt(r1)));
Kf=1+q*(Kt-1);
Kfs=1+qs*(Kts-1);
Kfl=1+ql*(Ktl-1);
Kfsl=1+qsl*(Ktsl-1);

% Calculating the for each gear to sit on

% Checking at Root of 2nd Gear
kb=(d/0.3)^-0.107;
Sen=Seg*ka*kb*ke;
A=sqrt(4*(Kf*Mroot)^2);
B=sqrt(3*(Kfs*Tm2)^2);
n=@(dq)((pi*dq^3)/16)*((A/(Sen*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n,0.5, 1.75,0.000000001,10000);

% Checking at pinion
kbg=(dg/0.3)^-0.107;
Seng=Seg*ka*kbg*ke;
A3=sqrt(4*(Mtot2)^2);
B3=sqrt(3*(Tm2)^2);
ng=@(dg)((pi*dg^3)/16)*((A3/(Seng*10^3))+(B3/(Sut*10^3)))^-1)-FOS;
dg=bisect(ng,0.5, 3,0.000000001,10000);

% Checking at root of first Gear
kbl=(d1/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A1=sqrt(4*(Kf1*Mroot1)^2);
B1=sqrt(3*(Kfs1*Tm2)^2);
n1=@(d1)((pi*d1^3)/16)*((A1/(Senl*10^3))+(B1/(Sut*10^3)))^-1)-FOS;
d1=bisect(n1,0.5, 3,0.000000001,10000);

% Recalculating for roots
r=d/10;
r1=d1/10;
q=1/(1+(neuber/sqrt(r)));
qs=1/(1+(neuberT/sqrt(r)));
ql=1/(1+(neuber/sqrt(r1)));
qsl=1/(1+(neuberT/sqrt(r1)));
Kf=1+q*(Kt-1);
Kfs=1+qs*(Kts-1);
Kfl=1+ql*(Ktl-1);
Kfsl=1+qsl*(Ktsl-1);

% Second Root calculation
A=sqrt(4*(Kf*Mroot)^2);
B=sqrt(3*(Kfs*Tm2)^2);
kb=(d/0.3)^-0.107;
Sen=Seg*ka*kb*ke;
n=@(dq)((pi*dq^3)/16)*((A/(Sen*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n,0.5, 1.75,0.000000001,10000);

% Sencond Gear calculation
kbg=(dg/0.3)^-0.107;

```

```

Seng=Seg*ka*kgb*ke;
A3=sqrt(4*(Mtot2)^2);
B3=sqrt(3*(Tm2)^2);
ng=@(dg)((pi*dg^3)/16)*((A3/(Seng*10^3))+(B3/(Sut*10^3)))^-1)-FOS;
dg=bisect(ng,0.5, 3,0.000000001,10000);

% First root Calculation
kbl=(d1/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A1=sqrt(4*(Kf1*Mroot1)^2);
B1=sqrt(3*(Kfs1*Tm2)^2);
n1=@(d1)((pi*d1^3)/16)*((A1/(Sen1*10^3))+(B1/(Sut*10^3)))^-1)-FOS;
d1=bisect(n1,0.5, 3,0.000000001,10000);

% Recalculating geometric stress
r=d/10;
r1=d1/10;
q=1/(1+(neuber/sqrt(r)));
qs=1/(1+(neuberT/sqrt(r)));
q1=1/(1+(neuber/sqrt(r1)));
qs1=1/(1+(neuberT/sqrt(r1)));
Kf=1+q*(Kt-1);
Kfs=1+qs*(Kts-1);
Kf1=1+q1*(Kt1-1);
Kfs1=1+qs1*(Kts1-1);

% Recalculating at second root
A=sqrt(4*(Kf*Mroot)^2);
B=sqrt(3*(Kfs*Tm2)^2);
kb=(d/0.3)^-0.107;
Sen=Seg*ka*kb*ke;
n=@(dq)((pi*dq^3)/16)*((A/(Sen*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n,0.5, 1.75,0.000000001,10000);

% Recalculating at second gear
kgb=(dg/0.3)^-0.107;
Seng=Seg*ka*kgb*ke;
A3=sqrt(4*(Mtot2)^2);
B3=sqrt(3*(Tm2)^2);
ng=@(dg)((pi*dg^3)/16)*((A3/(Seng*10^3))+(B3/(Sut*10^3)))^-1)-FOS;
dg=bisect(ng,0.5, 3,0.000000001,10000);

% Recalculating at first root
kbl=(d1/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A1=sqrt(4*(Kf1*Mroot1)^2);
B1=sqrt(3*(Kfs1*Tm2)^2);
n1=@(d1)((pi*d1^3)/16)*((A1/(Sen1*10^3))+(B1/(Sut*10^3)))^-1)-FOS;
d1=bisect(n1,0.5, 3,0.000000001,10000);

% Final diameter check for root
d=ceil(d*8)/8;
d1=ceil(d1*8)/8;

```

```

D=ceil((d*1.2)*8)/8;
D1st=ceil((d*5/3)*8)/8;
r=d/10;
r1=d1/10;
kb=(d/0.3)^-0.107;
kbl=(d1/0.3)^-0.107;
Sen=Seg*ka*kb*ke;
Sen1=Seg*ka*kbl*ke;

q=1/(1+(neuber/sqrt(r)));
qs=1/(1+(neuberT/sqrt(r)));
q1=1/(1+(neuber/sqrt(r1)));
qs1=1/(1+(neuberT/sqrt(r1)));
Kf=1+q*(Kt-1);
Kfs=1+qs*(Kts-1);
Kf1=1+q1*(Kt1-1);
Kfs1=1+qs1*(Kts1-1);

% Final FOS check for diameter at second root
A=sqrt(4*(Kf*Mroot)^2);
B=sqrt(3*(Kfs*Tm2)^2);
n=((pi*d^3)/16)*((A/(Sen*10^3))+(B/(Sut*10^3)))^-1);

% Final FOS check for diameter at first root
A1=sqrt(4*(Kf1*Mroot1)^2);
B1=sqrt(3*(Kfs1*Tm2)^2);
n1((((pi*d1^3)/16)*((A1/(Sen1*10^3))+(B1/(Sut*10^3)))^-1));

%Final check for for large diameter
dg=ceil(dg*8)/8;
kbg=(dg/0.3)^-0.107;
Seng=Seg*ka*kb*ke;
A3=sqrt(4*(Mtot2)^2);
B3=sqrt(3*(Tm2)^2);
ng=((pi*dg^3)/16)*((A3/(Seng*10^3))+(B3/(Sut*10^3)))^-1);

n=min(n,ng);
n=min(n,n1);
% Check for yielding
Ny2=(Sy*10^3)/((((32*Kf*Mroot)/(pi*d^3))^2)+(3*((16*Kfs*Tm2)/(pi*d^3))^2))^^(1/2));
Ng=(Sy*10^3)/((((32*Mtot2)/(pi*dg^3))^2)+(3*((16*Tm2)/(pi*dg^3))^2))^^(1/2));
Ny1=(Sy*10^3)/((((32*Kf1*Mroot1)/(pi*d1^3))^2)+(3*((16*Kfs1*Tm2)/(pi*d1^3))^2))^^(1/2));

Ny=min(Ny2,Ng);
Ny=min(Ny,Ny1);

disp(['The bore diameter for the first gear on the countershaft is ',
num2str(d1), ' inches'])
disp(['The bore diameter for the second gear on the Counter shaft is: ',
num2str(d), ' inches'])
disp(['The large diameter of Counter shaft is: ',num2str(max(D,dg)), '
inches'])

```

```

disp(['The radius of fillet is: ',num2str(r), ' inches'])
disp(['The Factor of Safety of Counter shaft is: ',num2str(n)])
disp(['Yielding Factor of Safety of the Counter shaft is: ', num2str(Ny)])

% Shaft 1

%Solving for Z reactions, shears, and moments
Tm1=Ft23*dp/2;
Rbz1=-Ft23*D1/Db1;
Raz1=-Ft23-Rbz1;
Vz1=Raz1;
Vz2=Vz1+Ft23;
Vz3=Vz2+Rbz1;
Mz1=Raz1*(D1);
Mz2=Mz1+Vz2*(D2);

%Solving for Y reactions, shears, and moments
Rby1=(Fr23*D1)/(Db1);
Ray1=-Rby1+Fr23;
Vy1=Ray1;
Vy2=Vy1-Fr23;
Vy3=Vy2+Rby1;
My1=Ray1*(D1);
My2=My1+Vy2*(D2);

%Solving for total moment
Mtot=sqrt(Mz1^2+My1^2);

% Initial Guesses and Marin Factors
ka=2*(Sut)^-0.217;
kb=0.9;
Se=Seg*ka*kb*ke;

A1=sqrt(4*(Mtot)^2);
B1=sqrt(3*(Tm1)^2);
d1=((16*FOS)/pi)*((A1/(Se*10^3))+(B1/(Sut*10^3)))^1/3;

%Solving for diameter of shaft 1
kbl=(d1/0.3)^-0.107;
Sen1=Seg*ka*kbl*ke;
A=sqrt(4*(Mtot)^2);
B=sqrt(3*(Tm1)^2);
n1=@(d1)((pi*d1^3)/16)*((A/(Sen1*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n1,0.25, 3,0.000000001,10000);

%Iterating with changes
kbl=(d/0.3)^-0.107;
Sen1=Seg*ka*kbl*ke;
A=sqrt(4*(Mtot)^2+3*(Kts*0)^2);
B=sqrt(4*(Kt*0)^2+3*(Tm1)^2);
n1=@(d1)((pi*d1^3)/16)*((A/(Sen1*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n1,0.25, 3,0.000000001,10000);

```

```

%Iterating again
kbl=(d/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A=sqrt(4*(Mtot)^2+3*(Kts*0)^2);
B=sqrt(4*(Kt*0)^2+3*(Tml)^2);
n1=@(d1)((pi*d1^3)/16)*((A/(Senl*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n1,0.25, 3,0.000000001,10000);

%One last Iteration
kbl=(d/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A=sqrt(4*(Mtot)^2+3*(Kts*0)^2);
B=sqrt(4*(Kt*0)^2+3*(Tml)^2);
n1=@(d1)((pi*d1^3)/16)*((A/(Senl*10^3))+(B/(Sut*10^3)))^-1)-FOS;
d=bisect(n1,0.25, 3,0.000000001,10000);

%Setting diameter to eights of an inch
d=ceil(d*8)/8;

kbl=(d/0.3)^-0.107;
Senl=Seg*ka*kbl*ke;
A=sqrt(4*(Mtot)^2+3*(Kts*0)^2);
B=sqrt(4*(Kt*0)^2+3*(Tml)^2);
n1=((pi*d^3)/16)*((A/(Senl*10^3))+(B/(Sut*10^3)))^-1);

% Check for yielding
Nyl=(Sy*10^3)/((((32*Mtot)/(pi*d^3))^2)+(3*((16*Tml)/(pi*d^3))^2)^(1/2));

disp(['The Diameter of 1st shaft is: ',num2str(d), ' inches'])
disp(['The Factor of Saftey of First shaft is: ',num2str(n1)])
disp(['Yeilding Factor of Safety of the First shaft is: ', num2str(Nyl)])

% Shaft 3

%Solving for Z reactions, shears, and moments
Tm3=Ft45*dg/2;
Rbz2=Ft45*D1/Db1;
Raz2=Ft45-Rbz2;
Vz4=Raz2;
Vz5=Vz4-Ft45;
Vz6=Vz5+Rbz2;
Mz3=Raz2*(D1);
Mz4=Mz3+Vz5*(D2);

%Solving for Y reactions, shears, and moments
Rby2=(Fr45*D1)/(Db1);
Ray2=Fr45-Rby2;
Vy4=Ray2;
Vy5=Vy4-Fr45;
Vy6=Vy5+Rby2;
My3=Ray2*(D1);
My4=My3+Vy5*(D2);

```

```

%Solving for total moment
Mtot3=sqrt (Mz3^2+My3^2);

kb=0.9; % guess of kb value
Se=Seg*ka*kb*ke; % initial Se with guesses

A=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
d3= ((16*FOS)/pi)* ((A/ (Se*10^3)) + (B/ (Se*10^3)))^1/3;

%Solving for diameter of shaft 3
kb3=(d3/0.3)^-0.107;
Sen3=Seg*ka*kb3*ke;
A3=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B3=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
n3=@(d1) (((pi*d1^3)/16)* ((A3/ (Sen3*10^3)) + (B3/ (Sut*10^3)))^-1))-FOS;
d3=bisect(n3,0.25,3,0.000000001,10000);

%Iterating with changes
kb3=(d3/0.3)^-0.107;
Sen3=Seg*ka*kb3*ke;
A3=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B3=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
n3=@(d1) (((pi*d1^3)/16)* ((A3/ (Sen3*10^3)) + (B3/ (Sut*10^3)))^-1))-FOS;
d3=bisect(n3,0.25,3,0.000000001,10000);

%Iterating again
kb3=(d3/0.3)^-0.107;
Sen3=Seg*ka*kb3*ke;
A3=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B3=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
n3=@(d1) (((pi*d1^3)/16)* ((A3/ (Sen3*10^3)) + (B3/ (Sut*10^3)))^-1))-FOS;
d3=bisect(n3,0.25,3,0.000000001,10000);

%One last Iteration
kb3=(d3/0.3)^-0.107;
Sen3=Seg*ka*kb3*ke;
A3=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B3=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
n3=@(d1) (((pi*d1^3)/16)* ((A3/ (Sen3*10^3)) + (B3/ (Sut*10^3)))^-1))-FOS;
d3=bisect(n3,0.25,3,0.000000001,10000);

%Setting diameter to eights of an inch
d3=ceil(d3*8)/8;

kb3=(d3/0.3)^-0.107;
Sen3=Seg*ka*kb3*ke;
A3=sqrt (4* (Mtot3)^2+3* (Kts*0)^2);
B3=sqrt (4* (Kt*0)^2+3* (Tm3)^2);
n3= ((pi*d3^3)/16)* ((A3/ (Sen3*10^3)) + (B3/ (Sut*10^3)))^-1);

% Check for yielding

```

```

Ny3=(Sy*10^3)/((((32*Mtot3)/(pi*d3^3))^2)+(3*((16*Tm3)/(
(pi*d3^3))^2))^(1/2));

disp(['The Diameter of 3rd shaft is: ',num2str(d3), ' inches'])
disp(['The Factor of Safety of Third shaft is: ',num2str(n3)])
disp(['Yielding Factor of Safety of the First shaft is: ', num2str(Ny3)])

% Bearing Info
AppF=1.2;
Rel=0.99;
a=3/10;
Ophr=5000;
X0=0.02;
Theta=4.459;
b=1.483;
Lr=10^6;
Bot=X0+((Theta-X0)*(log(1/Rel))^(1/b));

% Shaft 1

R1=sqrt(Rax1^2+Ray1^2);
Plrpm=2000;
LDP1=Plrpm*Ophr*60;
XDP1=LDP1/Lr;
C10P1=((AppF*R1)*(XDP1/Bot)^(a));

% Countershaft

Ra2=sqrt(Rax2^2+Raxy^2);
Glrpm=Plrpm*30/100;
LDG1=Glrpm*Ophr*60;
XDG1=LDG1/Lr;
C10G1=((AppF*Ra2)*(XDG1/Bot)^(a));

Rb2=sqrt(Rbx2^2+Rbxy^2);
P2rpm=Glrpm;
LDP2=P2rpm*Ophr*60;
XDP2=LDP2/Lr;
C10P2=((AppF*Rb2)*(XDP2/Bot)^(a));

% Shaft 3

R3=sqrt(Raz2^2+Ray2^2);
G2rpm=Glrpm*30/100;
LDG2=G2rpm*Ophr*60;
XDG2=LDG2/Lr;
C10G2=((AppF*R3)*(XDG2/Bot)^(a));

disp(['The bearings for shaft 1 need a C10 of : ', num2str(ceil(C10P1*10)/
10), ' lbf'])
disp(['The bearing near the Input gear for shaft 2 needs a C10 of : ',
num2str(ceil(C10G1*10)/10), ' lbf'])
disp(['The bearing far from the Input gear for shaft 2 needs a C10 of : ',

```

```

num2str(ceil(C10P2*10)/10), ' lbf'])
disp(['The bearings for shaft 3 need a C10 of : ', num2str(ceil(C10G2*10)/
10), ' lbf'])

% Finding force applied onto each screw

Tin=Ft23*dp/2;
Tout=Ft45*Dg/2;
Thouse=Tout+Tin;

% Distances

L=Dbc;
H=(Dg/2)+(dp/2)+2*((Dg/2)+(1/P))+0.3;
DTS1=sqrt((L/2)^2+(H/2)^2);
DTS2=H/2;
DTS3=L/2;
DTtot=(4*(DTS1^2)+2*(DTS2^2)+2*(DTS3^2));

%Torque on each screw
TPS1=Thouse*DTS1/DTtot;
TPS2=Thouse*DTS2/DTtot;
TPS3=Thouse*DTS3/DTtot;

disp(['The 4 screws on the corners must be able to resist ', num2str(TPS1),
' lbs of shear force'])
disp(['The left and right screws must be able to resist ', num2str(TPS2), '
lbs of shear force'])
disp(['The top and bottom screws must be able to resist ', num2str(TPS3), '
lbs of shear force'])

Number of teeth on the pinion is: 30 Teeth
Number of teeth on the gear is: 100 Teeth
Diametral Pitch is: 10 Teeth per inch
The circular pitch is: 0.31416 inches
Pinion Bending Stress is: 6295.7399 PSI
Bending FOS is: 4.8277
Pinion Contact Stress is: 54035.8867 PSI
Contact FOS is: 1.846
Pinion Bending Stress for our 2nd gearset is: 17808.9865 PSI
Bending FOS for our 2nd gearset is: 1.7428
Pinion Contact Stress for our 2nd gearset is: 90882.13 PSI
Contact FOS for our 2nd gearset is: 1.1278
The bore diameter for the first gear on the countershaft is 0.75 inches
The bore diameter for the second gear on the Counter shaft is: 1 inches
The large diameter of Counter shaft is: 1.25 inches
The radius of fillet is: 0.1 inches
The Factor of Saftey of Counter shaft is: 1.5
Yeilding Factor of Safety of the Counter shaft is: 3.5837
The Diameter of 1st shaft is: 0.5 inches
The Factor of Saftey of First shaft is: 1.7811
Yeilding Factor of Safety of the First shaft is: 4.4641
The Diameter of 3rd shaft is: 0.75 inches
The Factor of Saftey of Third shaft is: 2.1002

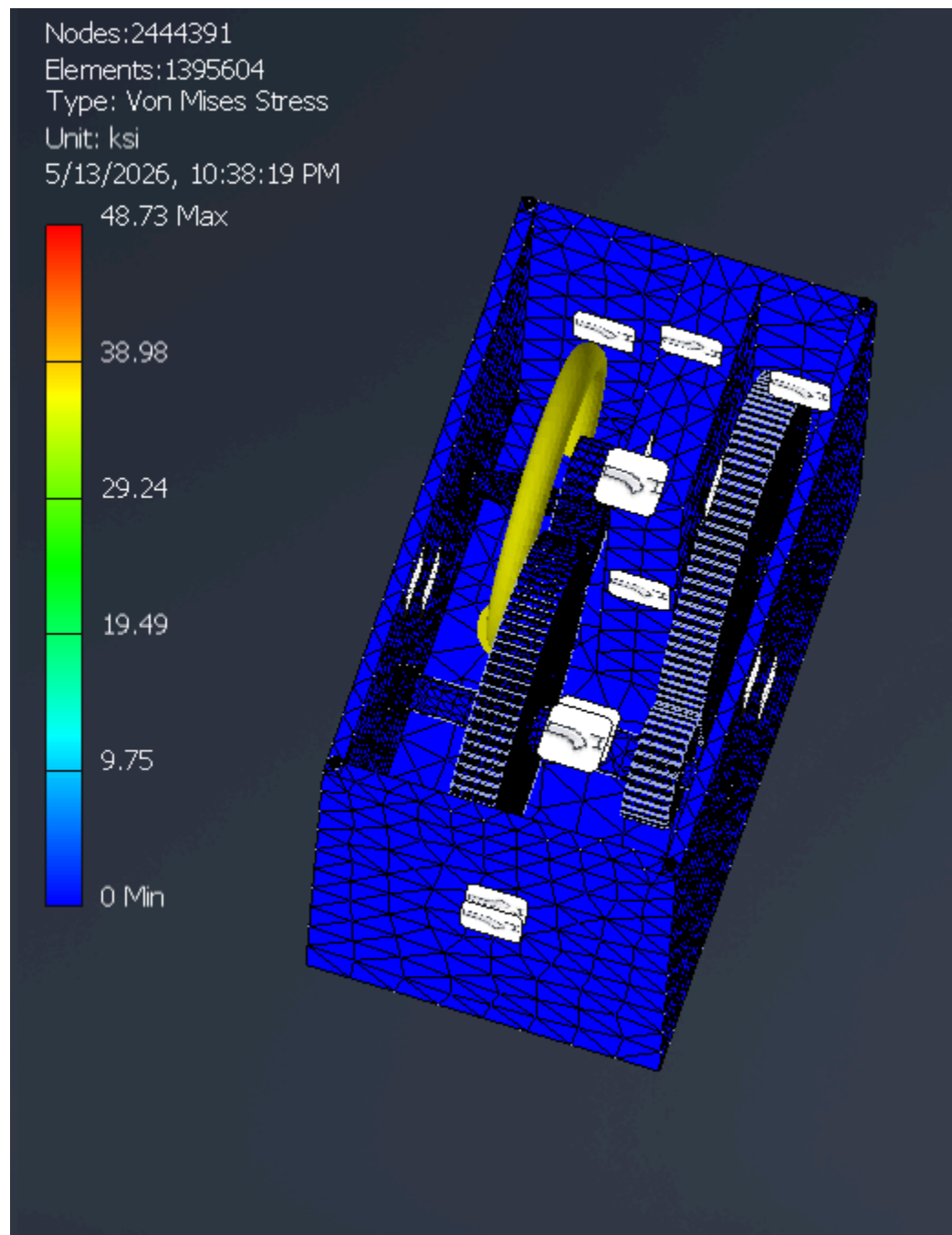
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Yeilding Factor of Safety of the First shaft is: 6.6064
The bearings for shaft 1 need a C10 of : 864.4 lbf
The bearing near the Input gear for shaft 2 needs a C10 of : 986.4 lbf
The bearing far from the Input gear for shaft 2 needs a C10 of : 2330.2 lbf
The bearings for shaft 3 need a C10 of : 1399.1 lbf
The 4 screws on the corners must be able to resist 38.7012 lbs of shear force
The left and right screws must be able to resist 33.3579 lbs of shear force
The top and bottom screws must be able to resist 19.6223 lbs of shear force

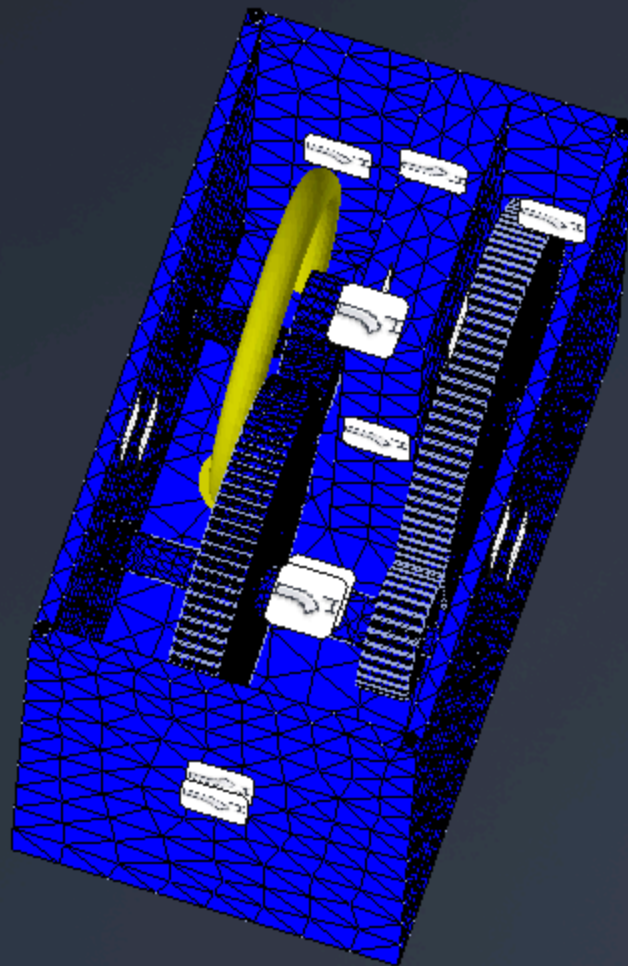
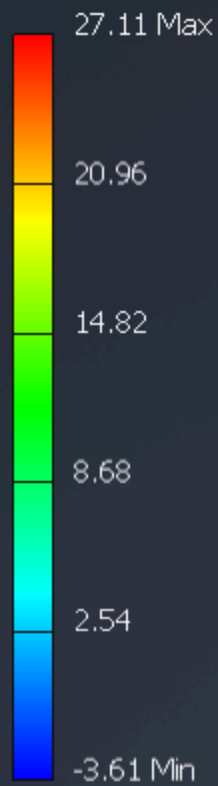
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Appendix B

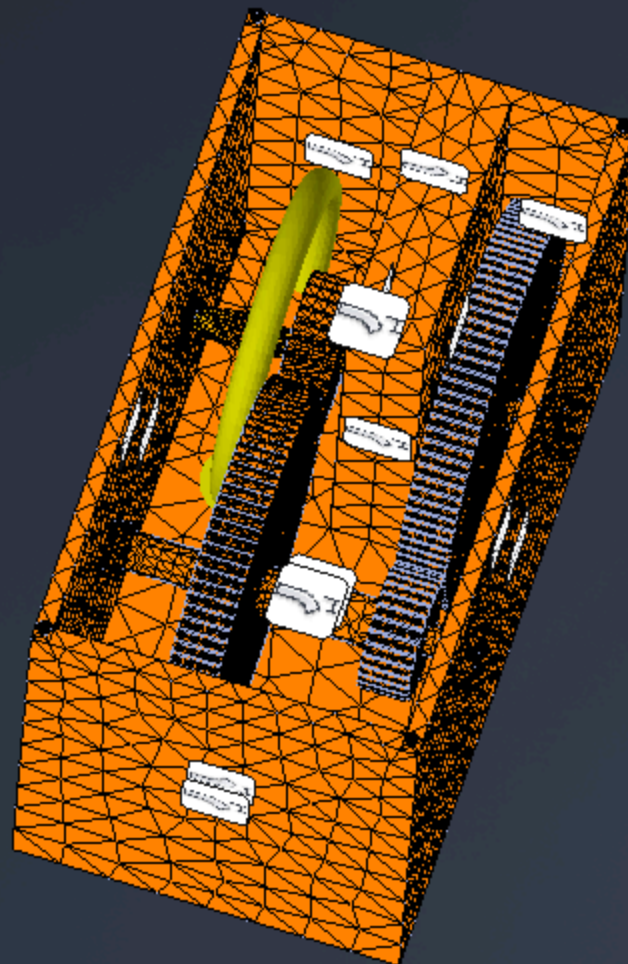
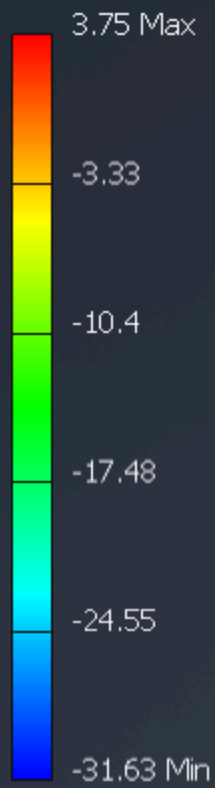
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Elements:1395604
Type: 1st Principal Stress
Unit: ksi
5/13/2026, 10:38:44 PM



Nodes:2444391
Elements:1395604
Type: 3rd Principal Stress
Unit: ksi
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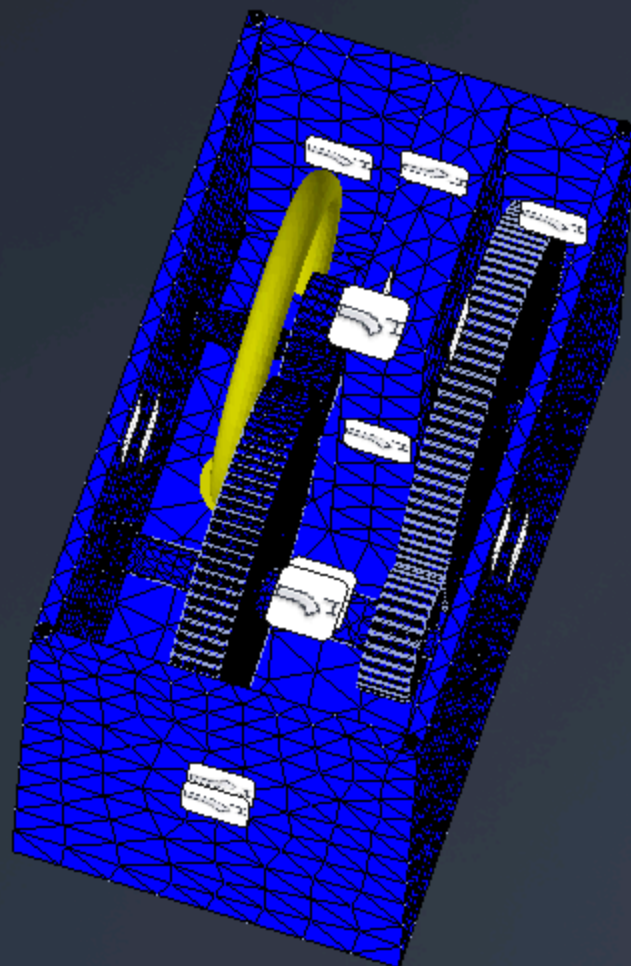
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Elements:1395604

Type: Safety Factor

Unit: ul

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Nodes:2444391

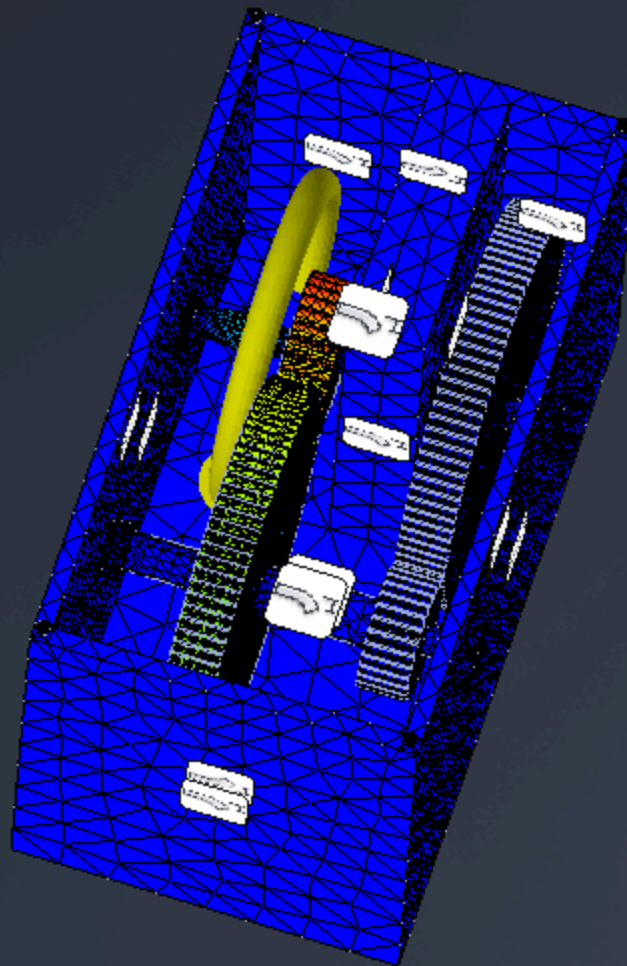
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Type: Displacement

Unit: in

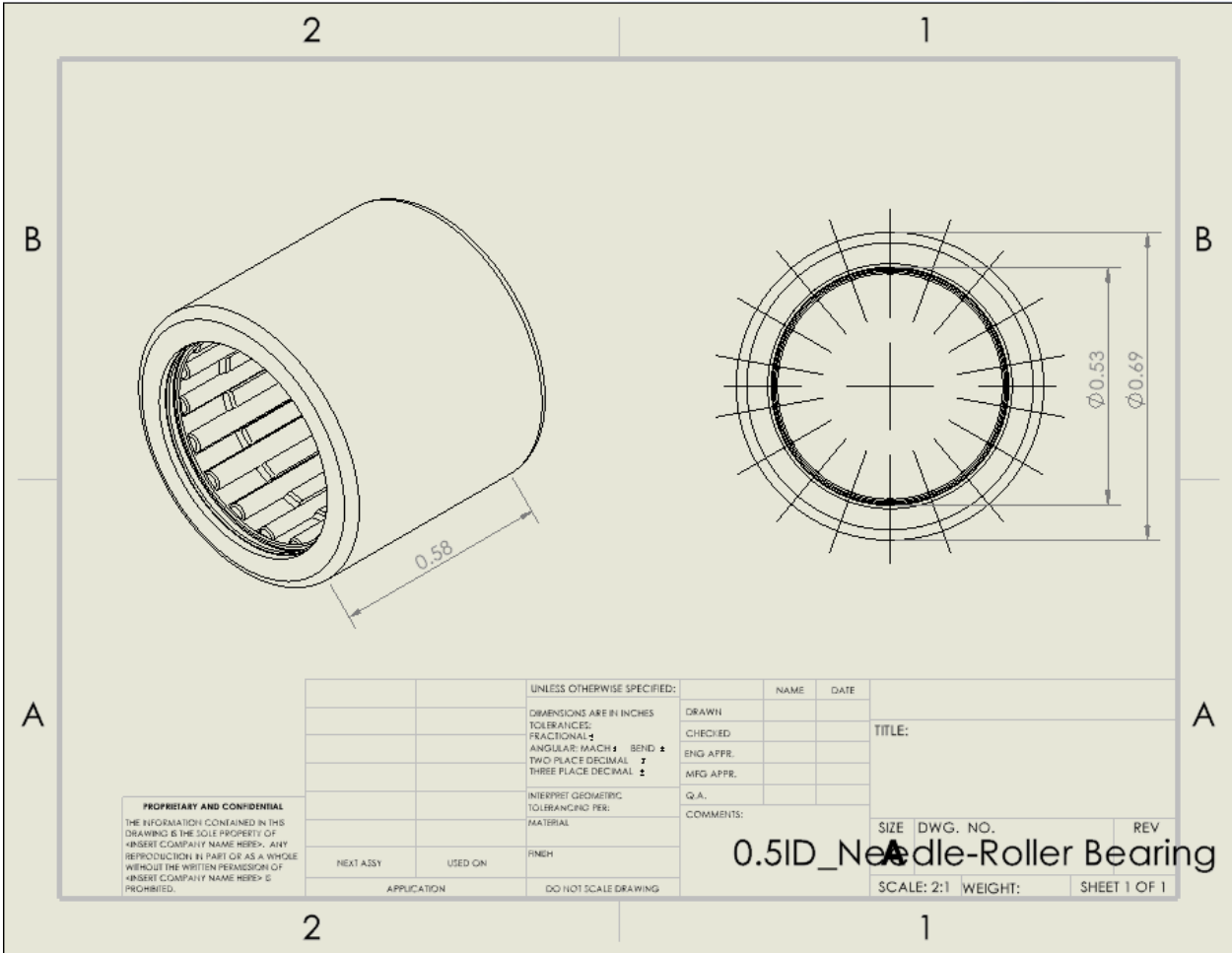
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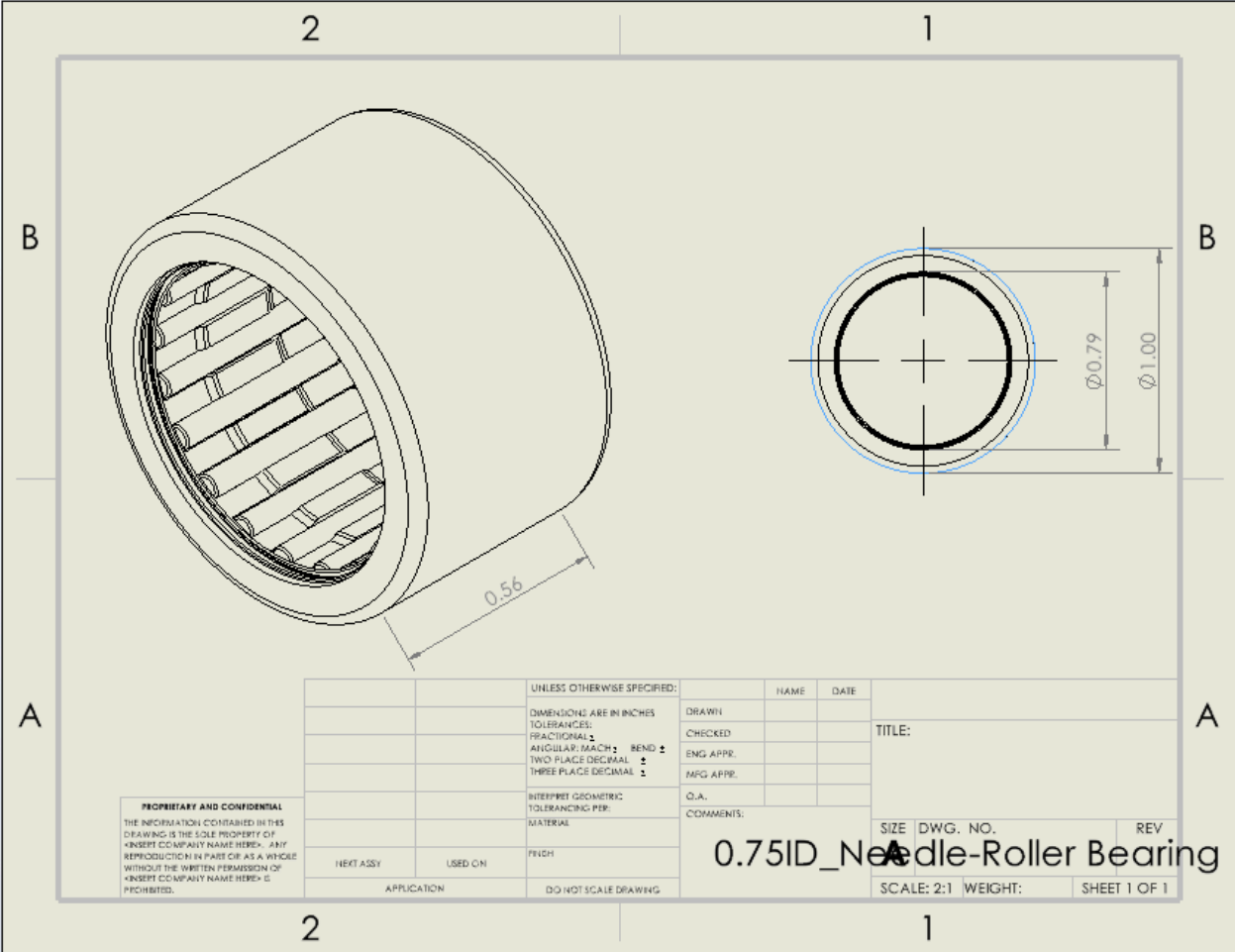
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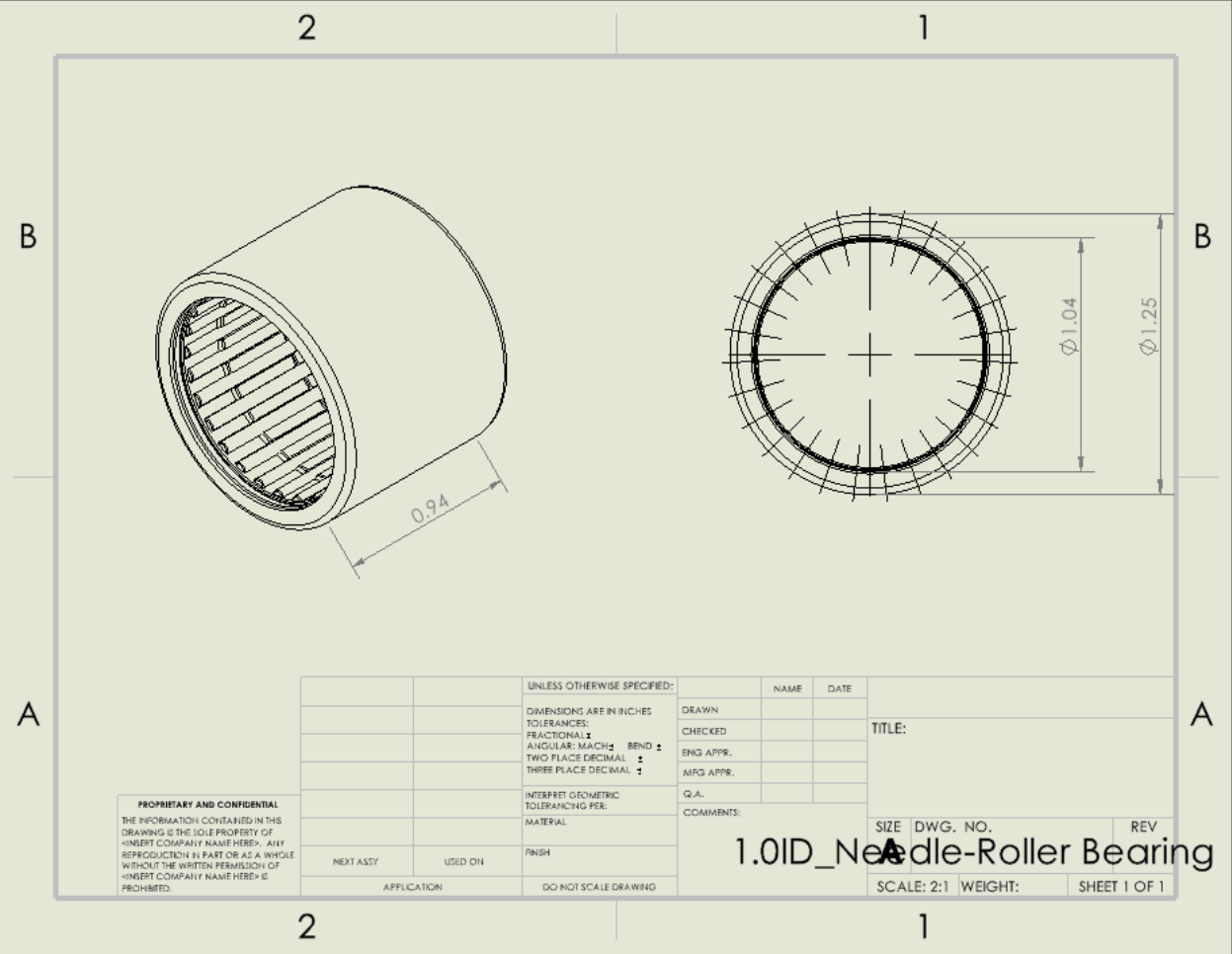


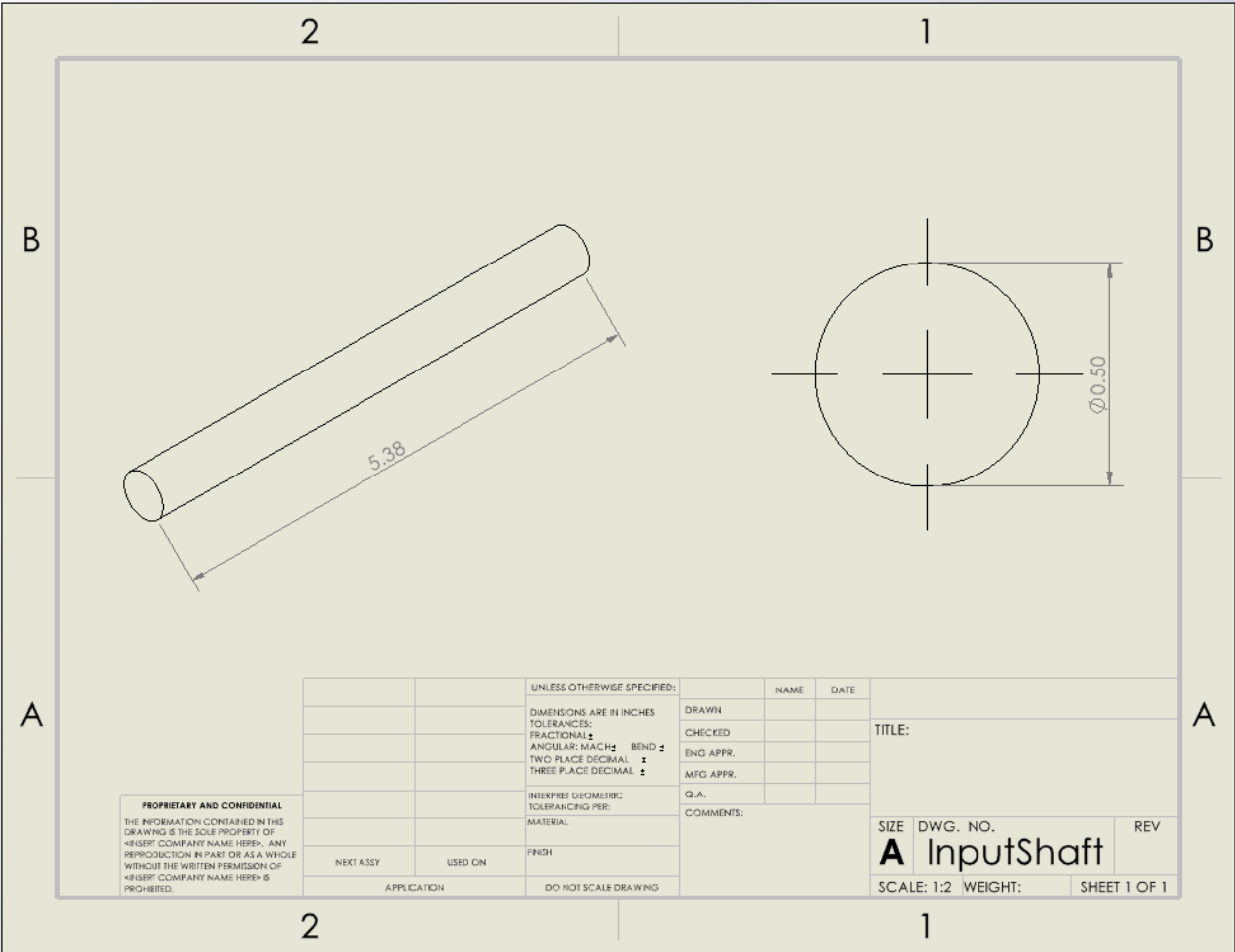
Appendix C

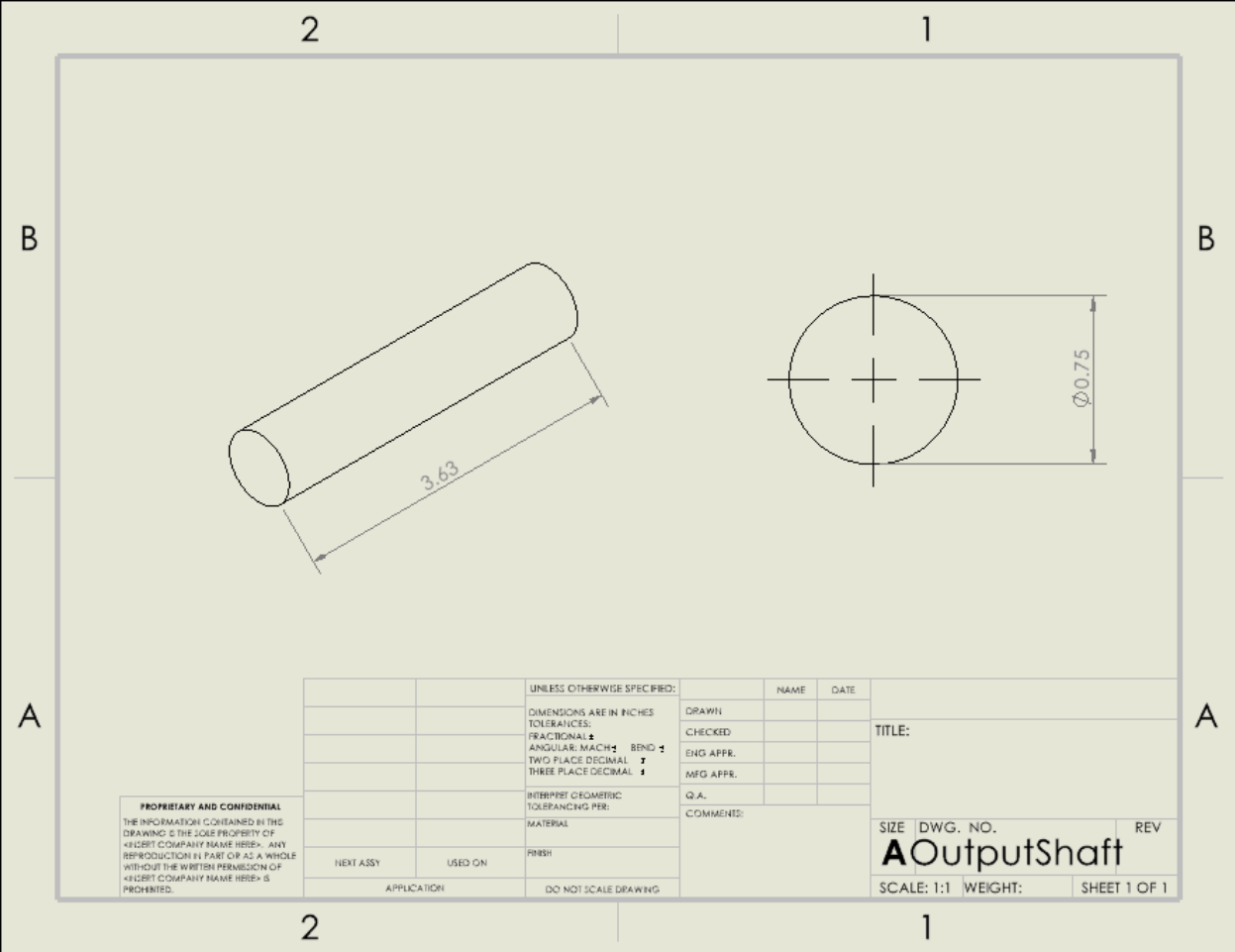
SW DRAWINGS

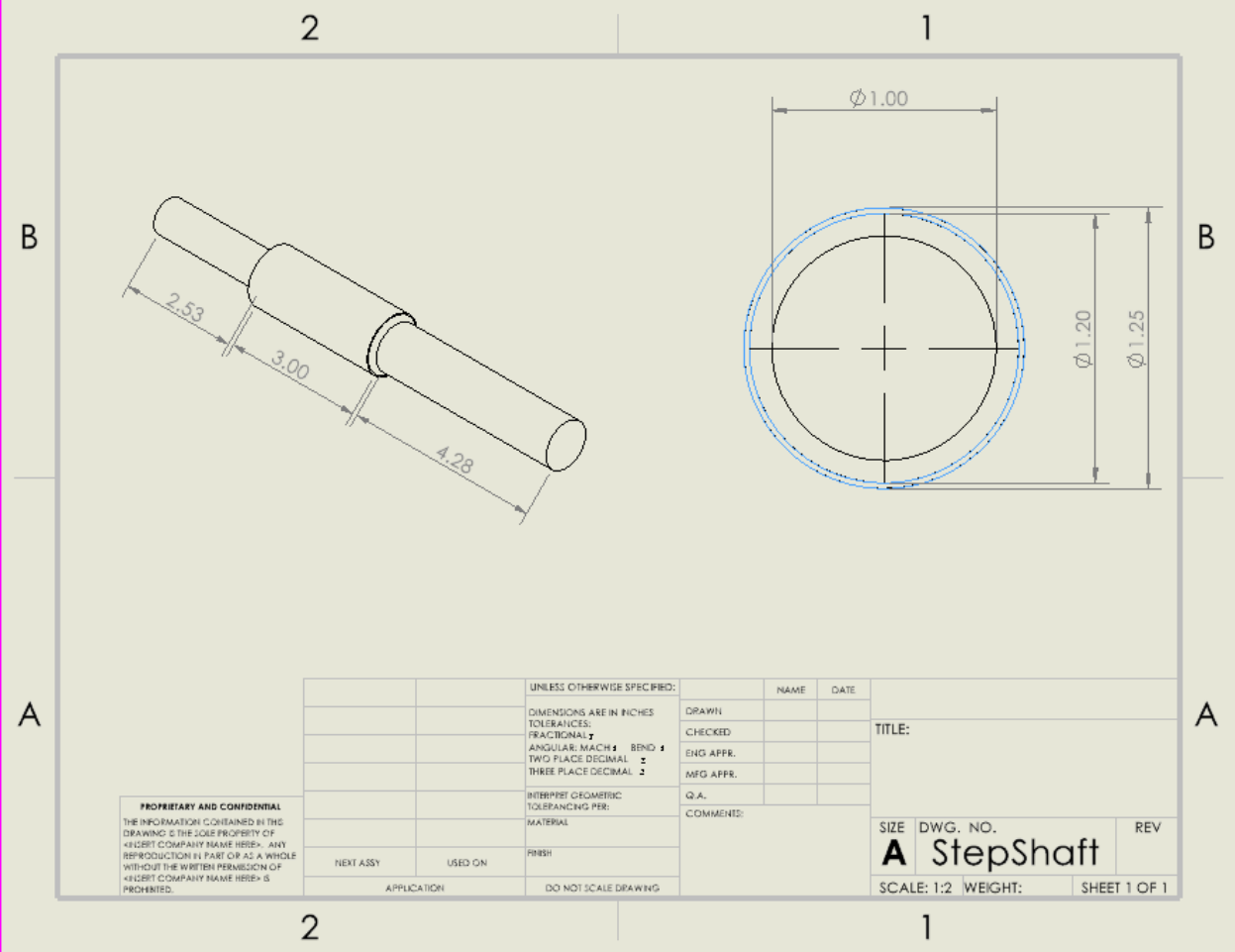


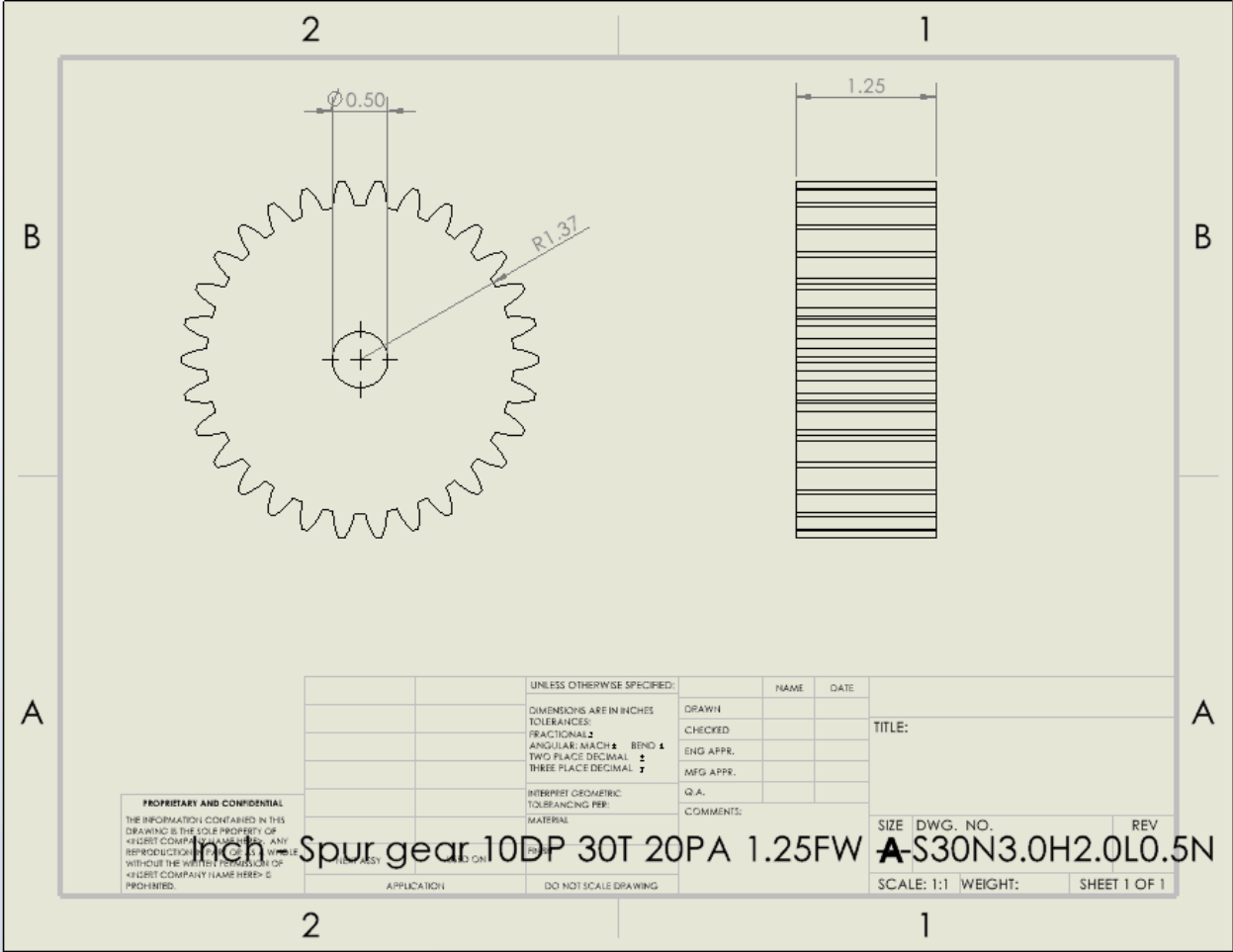


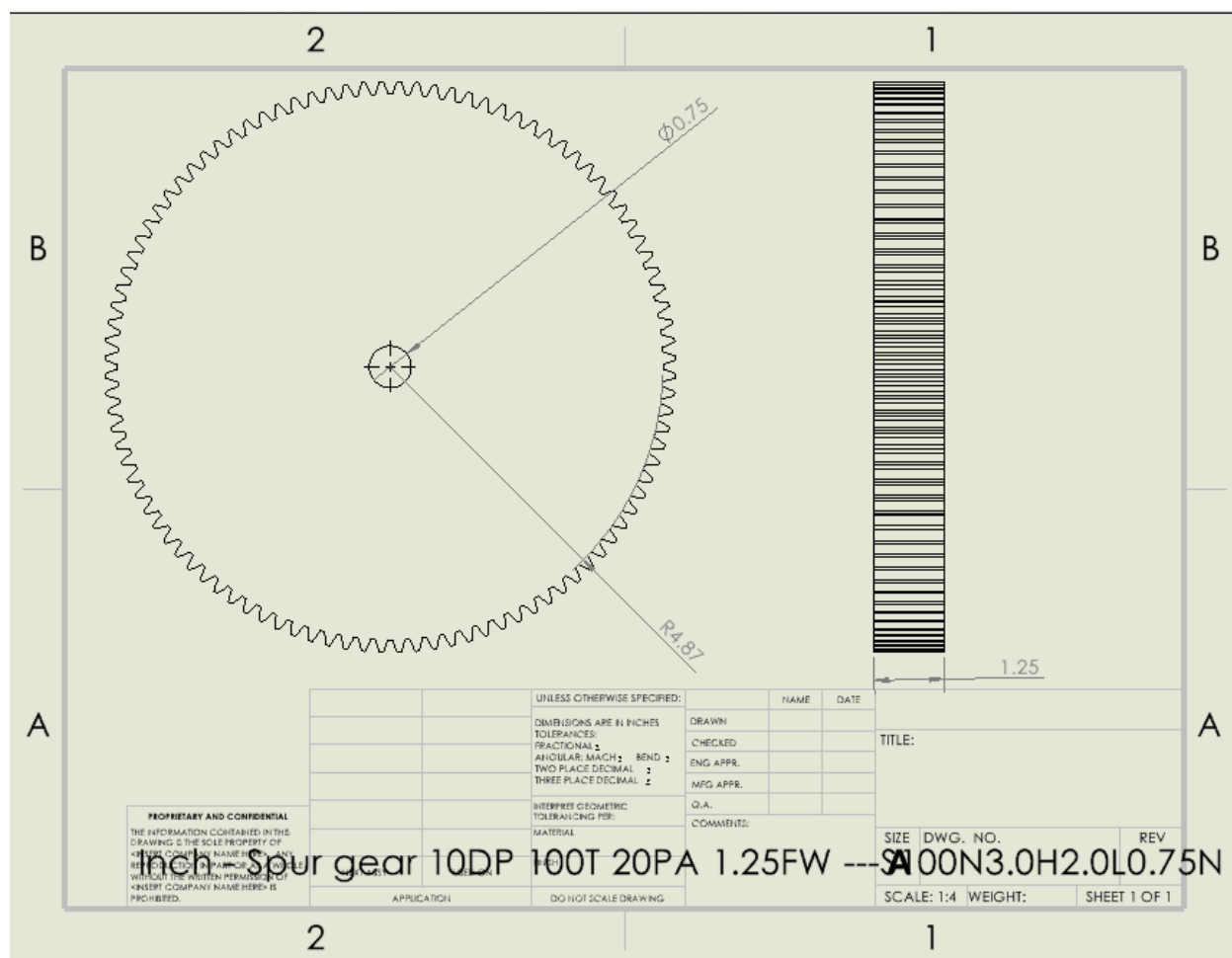


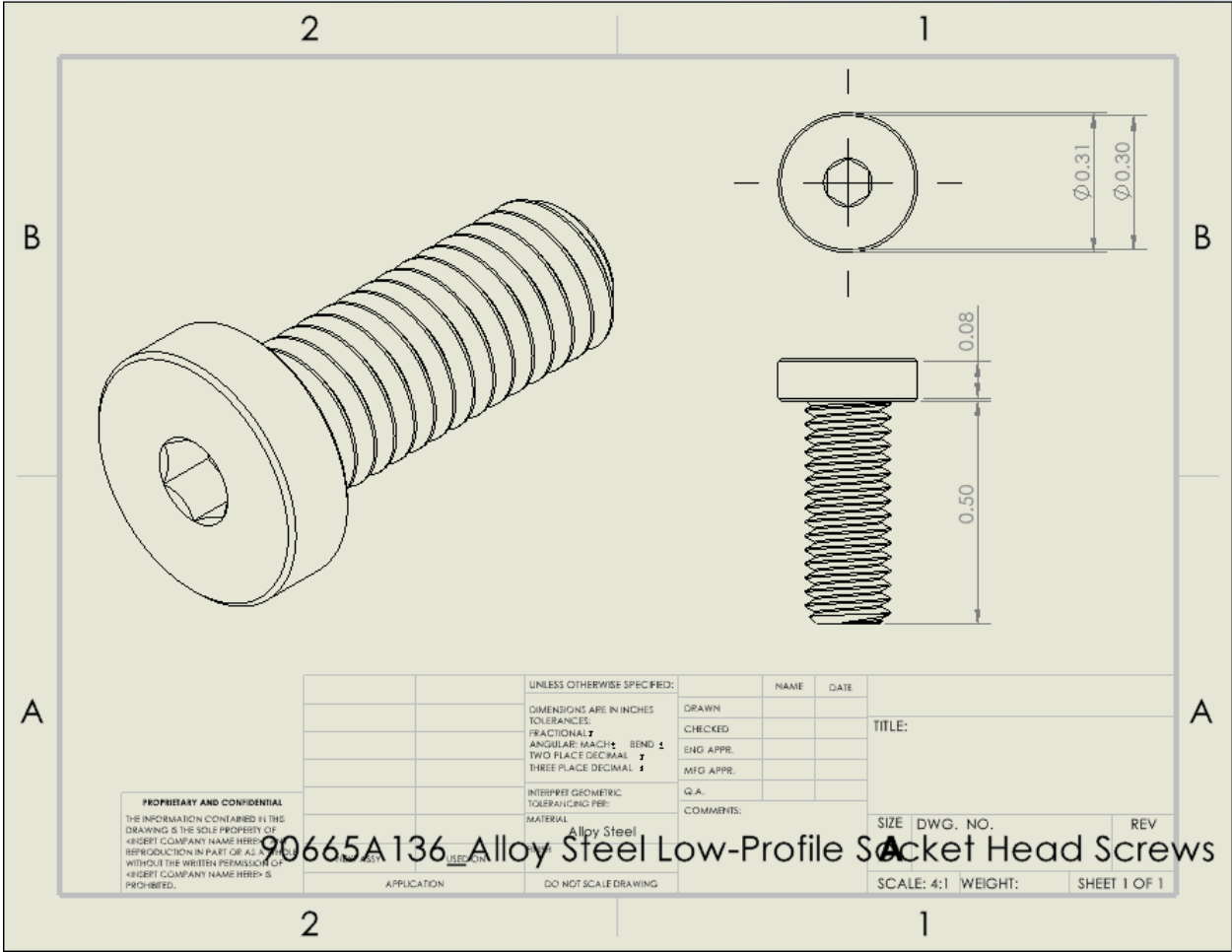












Appendix D

MATERIALS



Boston Gear (RRX)
YF100
Spur Gear, 10 DP, 20 PA, 100 Teeth, Cast Iron
MI ITEM # 00371921 | MFR # YF100

Availability: 🟡 Factory Order

🚚 **Shipping**
Shipping time may vary. Motion rep will advise.

Price **\$613.64** /each



Boston Gear (RRX)
YF30A
Spur Gear, 10 DP, 20 PA, 30 Teeth, Steel
MI ITEM # 00371890 | MFR # YF30A

Availability: 🟢 In stock

🚚 **Shipping**
[Get estimated arrival date](#) ⓘ

Price **\$210.14** /each

Made-to-Order 1045 Carbon Steel Rods



When an off-the-shelf size won't do, we'll get you the exact dimensions you need. Stronger than low-carbon steel with equally good machinability, 1045 carbon steel is widely used for bolts, studs, and shafts.

📄 Certificates with a traceable lot number are available for these products. Download certificates from **ORDER HISTORY** after your order ships.

Dia.	Dia. Tolerance	Choose a Length	Fabrication	Yield Strength, psi	Hardness	Specs. Met		Per Inch
1/2"	-0.003" to 0"	13" to 90"	Cold Worked	77,000	Rockwell B90 (Medium)	ASTM A108	7330N13	\$0.64
3/4"	-0.003" to 0"	13" to 90"	Cold Worked	77,000	Rockwell B90 (Medium)	ASTM A108	7330N15	1.00
1 1/4"	-0.003" to 0"	13" to 90"	Cold Worked	77,000	Rockwell B90 (Medium)	ASTM A108	7330N18	1.46

Needle-Roller Bearings



For Shaft Dia.	For Housing ID	Wd.	Radial Load Cap., lb.		Max. Rotation Speed, rpm	Lubrication	For Shaft Surface Smoothness, Ra, μ in	Temp. Range, ° F	Specs. Met	Each
			Dynamic	Static						
For Radial Load—Sealed on One Side with Steel Ring										
8 mm	12 mm	10 mm	620	590	17,700	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K84 11.45
12 mm	18 mm	14 mm	1,450	1,600	12,000	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K85 9.31
14 mm	20 mm	14 mm	1,600	1,900	10,500	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K86 13.14
15 mm	21 mm	14 mm	1,750	2,200	9,900	Lubricated	8	0 to 210	DIN 618	5905K572 12.28
15 mm	21 mm	18 mm	2,350	3,200	9,900	Lubricated	8	0 to 210	DIN 618	5905K573 13.75
16 mm	22 mm	14 mm	1,700	2,150	9,300	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K87 14.41
20 mm	26 mm	18 mm	2,850	4,500	7,600	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K88 11.48
22 mm	28 mm	18 mm	3,000	4,950	7,000	Lubricated	8	0 to 210	DIN 618	5905K576 18.11
25 mm	32 mm	18 mm	3,500	5,350	6,100	Lubricated	8	0 to 210	DIN 618	5905K577 15.88
30 mm	37 mm	18 mm	3,850	6,500	5,200	Lubricated	8	0 to 210	DIN 618	5905K578 21.52
For Radial Load—Sealed on Both Sides with Steel Ring										
1/4"	7/16"	7/16"	570	470	21,000	Lubricated	8	-20 to 210	—	5905K131 13.28
5/16"	1/2"	7/16"	650	590	17,500	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K583 13.07
3/8"	9/16"	7/16"	650	610	15,000	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K584 14.07
3/8"	9/16"	5/8"	1,300	1,450	15,000	Lubricated	8	-20 to 210	—	5905K132 11.70
1/2"	11/16"	7/16"	780	820	15,000	Lubricated	8	-25 to 225	—	5905K335 14.99
1/2"	11/16"	5/8"	1,550	2,000	11,700	Lubricated	8	-20 to 210	—	5905K133 12.14

Needle-Roller Bearings



For Shaft Dia.	For Housing ID	Wd.	Radial Load Cap., lb.		Max. Rotation Speed, rpm	Lubrication	For Shaft Surface Smoothness, Ra, µin	Temp. Range, ° F	Specs. Met	Each
			Dynamic	Static						
For Radial Load—Sealed on One Side with Steel Ring										
15 mm	21 mm	14 mm	1,750	2,200	9,900	Lubricated	8	0 to 210	DIN 618	5905K572 12.28
15 mm	21 mm	18 mm	2,350	3,200	9,900	Lubricated	8	0 to 210	DIN 618	5905K573 13.75
16 mm	22 mm	14 mm	1,700	2,150	9,300	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K87 14.41
20 mm	26 mm	18 mm	2,850	4,500	7,600	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K88 11.48
22 mm	28 mm	18 mm	3,000	4,950	7,000	Lubricated	8	0 to 210	DIN 618	5905K576 18.11
25 mm	32 mm	18 mm	3,500	5,350	6,100	Lubricated	8	0 to 210	DIN 618	5905K577 15.88
30 mm	37 mm	18 mm	3,850	6,500	5,200	Lubricated	8	0 to 210	DIN 618	5905K578 21.52
For Radial Load—Sealed on Both Sides with Steel Ring										
1/4"	7/16"	7/16"	570	470	21,000	Lubricated	8	-20 to 210	—	5905K131 13.28
5/16"	1/2"	7/16"	650	590	17,500	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K583 13.07
3/8"	9/16"	7/16"	650	610	15,000	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K584 14.07
3/8"	9/16"	5/8"	1,300	1,450	15,000	Lubricated	8	-20 to 210	—	5905K132 11.70
1/2"	11/16"	7/16"	780	820	15,000	Lubricated	8	-25 to 225	—	5905K335 14.99
1/2"	11/16"	5/8"	1,550	2,000	11,700	Lubricated	8	-20 to 210	—	5905K133 12.14
5/8"	13/16"	5/8"	1,800	2,700	9,600	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K585 14.04
5/8"	13/16"	3/4"	2,300	3,650	9,600	Lubricated	8	-20 to 210	—	5905K134 12.87
3/4"	1"	5/8"	2,200	2,800	7,800	Lubricated	8	-20 to 210	—	5905K135 12.87

Needle-Roller Bearings



		Radial Load Cap., lb.								
For Shaft Dia.	For Housing ID	Wd.	Dynamic	Static	Max. Rotation Speed, rpm	Lubrication	For Shaft Surface Smoothness, Ra, µin	Temp. Range, ° F	Specs. Met	Each
For Radial Load—Sealed on One Side with Steel Ring										
16 mm	22 mm	14 mm	1,700	2,150	9,300	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K87 14.41
20 mm	26 mm	18 mm	2,850	4,500	7,600	Lubricated	8	-20 to 210	DIN 618, ISO 3245	5905K88 11.48
22 mm	28 mm	18 mm	3,000	4,950	7,000	Lubricated	8	0 to 210	DIN 618	5905K576 18.11
25 mm	32 mm	18 mm	3,500	5,350	6,100	Lubricated	8	0 to 210	DIN 618	5905K577 15.88
30 mm	37 mm	18 mm	3,850	6,500	5,200	Lubricated	8	0 to 210	DIN 618	5905K578 21.52
For Radial Load—Sealed on Both Sides with Steel Ring										
1/4"	7/16"	7/16"	570	470	21,000	Lubricated	8	-20 to 210	—	5905K131 13.28
5/16"	1/2"	7/16"	650	590	17,500	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K583 13.07
3/8"	9/16"	7/16"	650	610	15,000	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K584 14.07
3/8"	9/16"	5/8"	1,300	1,450	15,000	Lubricated	8	-20 to 210	—	5905K132 11.70
1/2"	11/16"	7/16"	780	820	15,000	Lubricated	8	-25 to 225	—	5905K335 14.99
1/2"	11/16"	5/8"	1,550	2,000	11,700	Lubricated	8	-20 to 210	—	5905K133 12.14
5/8"	13/16"	5/8"	1,800	2,700	9,600	Lubricated	8	0 to 210	ISO 281, ISO 76	5905K585 14.04
5/8"	13/16"	3/4"	2,300	3,650	9,600	Lubricated	8	-20 to 210	—	5905K134 12.87
3/4"	1"	5/8"	2,200	2,800	7,800	Lubricated	8	-20 to 210	—	5905K135 12.87
3/4"	1"	3/4"	2,850	3,950	7,800	Lubricated	8	-20 to 210	—	5905K136 12.96
1"	1 1/4"	1"	4,050	6,800	6,100	Lubricated	8	-20 to 210	—	5905K137 16.71

Steel Socket Head Screws



Fully Threaded



Partially Threaded

Lg.	Head		Threading	Thread Spacing	Tensile Strength, psi	Mil. Spec.	Fastener Strength Grade/Class	Pkg. Qty.	Pkg.
	Dia.	Ht.							
Zinc-Plated Alloy Steel									
10-24									
1 1/8"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	25	90128A128 7.00
1 1/4"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	25	90128A225 6.29
1 1/2"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	25	90128A226 7.13
1 3/4"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	25	90128A227 8.67
2"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	25	90128A229 9.34
2 1/4"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	10	90128A231 10.98
2 1/2"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	10	90128A233 11.42
2 3/4"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	5	90128A235 7.03
3"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	5	90128A237 11.02
3 1/2"	5/16"	0.19"	Partially Threaded	Coarse	170,000	—	—	5	90128A238 9.88
10-32									
3/16"	5/16"	0.19"	Fully Threaded	Fine	170,000	—	—	50	90128A129 12.54
1/4"	5/16"	0.19"	Fully Threaded	Fine	170,000	—	—	50	90128A932 11.53
5/16"	5/16"	0.19"	Fully Threaded	Fine	170,000	—	—	50	90128A131 6.45
3/8"	5/16"	0.19"	Fully Threaded	Fine	170,000	—	—	50	90128A940 8.59
7/16"	5/16"	0.19"	Fully Threaded	Fine	170,000	—	—	25	90128A315 12.01
1/2"	5/16"	0.19"	Fully Threaded	Fine ⓘ	170,000	—	—	50	90128A942 8.80